

QUANTUM MECHANICS

Foundations and Applications

IISER BHOPAL · SAATHVIK BOGAM

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Contents

1	Fundamentals	3
1.1	Bra Ket Notation	3
1.1.1	Kets, Bras and Inner Products	3
1.1.2	Operators and Outer Products	5
1.1.3	Eigenvalues, Eigenkets and Observables	6
1.2	Measurement and Uncertainty	8
1.2.1	Measurement	8
1.2.2	Compatibility of Observables	9
1.2.3	Incompatibility of Observables	10
1.2.4	Uncertainty Relation	11
1.2.5	Unitary Operators and Change of Basis	12
1.3	Position and Momentum	14
1.3.1	Continuous Spectra	14
1.3.2	Position	14
1.3.3	Translation	15
1.3.4	Momentum	16
1.3.5	Wave Functions	18
1.3.6	Momentum and Position Operators	19
1.3.7	Generalization to Three Dimensions	21
1.4	Schrödinger Equation	21
1.4.1	Time Evolution	21
1.4.2	Schrödinger Equation	22
1.4.3	Energy Eigenkets	23
1.4.4	Schrödinger's Wave Equation	24
1.4.5	Interpretation of the Wave Function	25
2	One Dimensional Problems	26
2.0.1	Bound and Scattering States	26
2.0.2	Parity Operator	26
2.1	Infinite Square Well	28

2.2	Finite Square Well	30
2.3	Delta Function Well	33
2.4	Harmonic Oscillator	35
2.4.1	Power Series Solution	35
2.4.2	Dirac's Solution	37
2.5	Free Particle and Wave Packets	39
2.5.1	Free Particle	39
2.5.2	Wave Packets	40
2.6	Scattering States	41
2.6.1	Transmission and Reflection	41
2.6.2	Scattering and Transfer Matrices	42
2.6.3	Finite Square Well	44
2.6.4	Delta Function Well	45
2.6.5	Potential Step	46
3	Hydrogen and Angular Momentum	47
3.1	Hydrogen Atom	47
3.1.1	Radial Equation	47
3.1.2	Spherical Harmonics	48
3.1.3	Hydrogen Atom	49
3.2	Angular Momentum	50
3.2.1	Rotation Commutation	50
3.2.2	Rotation Operator	51
3.2.3	Generalized Angular Momentum	52
3.2.4	Eigensystem of Angular Momentum	53
3.2.5	Spin States and Spin 1/2	55
3.2.6	Eigenfunctions of Orbital Angular Momentum	57
3.2.7	Addition of Angular Momentum	58

1 FUNDAMENTALS

This section shall cover the fundamentals of Quantum Mechanics, starting with the vector formalism, the Schrödinger Equation and other concepts, so that one is ready to solve any problems they encounter in the Wave Mechanics part of these notes. I've chosen to take this approach as I believe that the most optimal way to study physics is to start by having a strong mathematical formalism. You will only require a knowledge of introductory Linear algebra, Integral and Differential calculus to start.

1.1 Bra Ket Notation

In Classical Mechanics, the movement of a system is dictated by its initial conditions and the forces acting on it. As long as you know all of these, you can predict any future that the system has. This is the concept of *trajectory*, it is the path that a particle traces over time, which can one obtain by solving Newton's Equation: $F = ma$. Quantum Mechanics is however, not so simple. We shall start with the concept of *the state*. The state is the entity that contains all information about the system, and by the language of Linear Algebra, we can extract the information out from this system. The state is a vector, specifically a *ket vector*, represented by $|\mathcal{S}\rangle$. The state ket lives in what is called a *Hilbert Space*, which is an abstract vector space represented by $\mathcal{H}$.

1.1.1 Kets, Bras and Inner Products

The ket is an abstract vector that lives in a vector space called the Hilbert space. A ket is represented by $|\cdot\rangle$. Being a vector, obviously it has the standard properties that a vector space possesses.

- The addition of two kets gives us another ket.

$$|\alpha\rangle + |\beta\rangle = |\gamma\rangle \quad (1.1)$$

- The product of a scalar and ket is a ket, if the constant is zero then the ket produced is the null ket.

$$c(|\alpha\rangle) = c|\alpha\rangle = |\alpha\rangle c \quad (1.2)$$

Then there is the *bra*, which is essentially a mirror of a ket.

Postulate 1.1: Dual Correspondence

For every ket $|\alpha\rangle$ in a Hilbert space $\mathcal{H}$, there exists a bra $\langle\alpha|$ in the Hilbert Dual space $\mathcal{H}_d$. The bra and ket are said to have a Dual Correspondence.

$$|\alpha\rangle \overset{DC}{\longleftrightarrow} \langle\alpha|$$

Bras work the same way as kets do, both are closed over addition and scalar multiplication.

In a finite dimensional vector space of N dimensions, kets and bras can be thought of as a column

and row matrix respectively. If in some basis, the ket is represented as a column matrix, then the dual bra of the ket is the *hermitian adjoint* of the ket. A hermitian adjoint is just the transpose conjugate of a matrix.

$$|\alpha\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \end{pmatrix} \xleftrightarrow{DC} \begin{pmatrix} \alpha_1^* & \alpha_2^* & \alpha_3^* & \dots \end{pmatrix} = \langle\alpha| \quad (1.3)$$

And in general,

$$a|\alpha\rangle + b|\beta\rangle \xleftrightarrow{DC} a^*\langle\alpha| + b^*\langle\beta| \quad (1.4)$$

The *inner product* is defined by the product of a bra to the left of the ket, and it is a complex scalar. You will also see the inner product to be referred to as a scalar product, but the inner product is just a generalization of the geometric scalar product.

$$\langle\alpha|\beta\rangle = \langle\alpha||\beta\rangle$$

Note that unlike the scalar product, $\langle\alpha|\beta\rangle \neq \langle\beta|\alpha\rangle$, but instead they are conjugates of each other, i.e.

$$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^* \quad (1.5)$$

Postulate 1.2: Positive Definite Metric

The inner product of a ket $|\alpha\rangle$ and its bra dual $\langle\alpha|$ is always real and positive and is called the norm, squared. It is zero if and only if the null ket is taken.

$$||\alpha||^2 = \langle\alpha|\alpha\rangle \geq 0$$

The inner product of $\langle\alpha|$ with a sum of kets $|\beta\rangle + |\gamma\rangle$ is the sum of the inner products, i.e.

$$\langle\alpha| [|\beta\rangle + |\gamma\rangle] = \langle\alpha|\beta\rangle + \langle\alpha|\gamma\rangle \quad (1.6)$$

The same applies to an inner product of a sum of bras and a ket.

$$[\langle\alpha| + \langle\beta|] |\gamma\rangle = \langle\alpha|\gamma\rangle + \langle\beta|\gamma\rangle \quad (1.7)$$

The inner product in an N dimensional space is defined as:

$$\langle\alpha|\beta\rangle = \begin{pmatrix} \alpha_1^* & \alpha_2^* & \dots & \alpha_n^* \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} = \alpha_1^*\beta_1 + \alpha_2^*\beta_2 + \dots + \alpha_n^*\beta_n \quad (1.8)$$

1.1.2 Operators and Outer Products

An operator is a map from one vector space to another, typically represented by $\hat{A}: V_1 \rightarrow V_2$. An operator acts on kets to produce another ket, and it is analogous to a linear transformation by a matrix. In fact, for a finite dimensional space with N dimensions, any operator can be represented as an $N \times N$ matrix. It being like a matrix, is also a linear entity, and therefore:

$$\hat{A}[a|\alpha\rangle + b|\beta\rangle] = a\hat{A}|\alpha\rangle + b\hat{A}|\beta\rangle \quad (1.9)$$

Operators can be multiplied, and the multiplication is associative and distributive.

$$\hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C} = \hat{A}\hat{B}\hat{C} \quad (1.10)$$

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C} \quad (1.11)$$

However, it may or may not always be commutative. The non-commutativity of operators is so important in Quantum Mechanics that we have two quantities that measure the commutativity of two operators: *the commutator and anti-commutator*, defined as follows, respectively:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \quad (1.12)$$

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A} \quad (1.13)$$

Commutators typically follow two important identities:

Leibniz Identity

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} - \hat{B}[\hat{A}, \hat{C}] \quad (1.14)$$

Jacobi Identity

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \quad (1.15)$$

The dual correspondent of $\hat{A}|\alpha\rangle$ is $\langle\alpha|\hat{A}^\dagger$ where $\hat{A}^\dagger$ is the hermitian conjugate of $\hat{A}$. An operator such that $\hat{A} = \hat{A}^\dagger$, where both have the same domain, is called a *hermitian operator*. For any two operators $\hat{A}$ and $\hat{B}$,

$$\langle\alpha|\hat{A}|\beta\rangle = [\langle\beta|\hat{A}^\dagger|\alpha\rangle]^* \quad (1.16)$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger \quad (1.17)$$

The *outer product* of a ket and bra is defined with the ket to the left and the bra to the right, and it is an operator, since it has the property of "rotating" a ket in the direction of the ket in

the outer product.

$$[|\alpha\rangle\langle\beta|]|\gamma\rangle = |\alpha\rangle\langle\beta|\gamma\rangle \quad (1.18)$$

In an N dimensional space, the outer product is as follows:

$$|\alpha\rangle\langle\beta| = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \begin{pmatrix} \beta_1^* & \beta_2^* & \dots & \beta_n^* \end{pmatrix} = \begin{pmatrix} \alpha_1\beta_1^* & \alpha_1\beta_2^* & \dots & \alpha_1\beta_n^* \\ \alpha_2\beta_1^* & \alpha_2\beta_2^* & \dots & \alpha_2\beta_n^* \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n\beta_1^* & \alpha_n\beta_2^* & \dots & \alpha_n\beta_n^* \end{pmatrix} \quad (1.19)$$

If we have a vector $|\gamma\rangle$ that is normalized, then the outer product $|\gamma\rangle\langle\gamma|$ is called a projection operator, represented by Λ_γ .

$$\Lambda_\gamma = |\gamma\rangle\langle\gamma| \quad (1.20)$$

1.1.3 Eigenvalues, Eigenkets and Observables

The eigenvalue equation of an operator $\hat{A}$, with eigenvalue A and corresponding eigenket $|A\rangle$. Typically, we only consider normalized eigenkets i.e. $\langle A|A\rangle = 1$ since $|A\rangle$ and $c|A\rangle$ represent the same state.

$$\hat{A}|A\rangle = A|A\rangle \quad (1.21)$$

Hermitian operators have a big role to play in this: they represent an experimental *observable*, be it position, momentum or energy, all these are represented by a Hermitian operator. This comes from the fact that the eigenvalues of Hermitian operators are only real.

Reality of Eigenvalues and Orthogonality of Eigenkets of Hermitian Operators

Consider two eigenvalues A and A' of a Hermitian operator $\hat{A}$ with corresponding eigenkets $|A\rangle$ and $|A'\rangle$.

$$\hat{A}|A\rangle = A|A\rangle \quad \hat{A}|A'\rangle = A'|A'\rangle \quad (1.22)$$

Applying the bra $\langle A'|$ to left in the first equation,

$$\langle A'|\hat{A}|A\rangle = A\langle A'|A\rangle \implies \langle A|\hat{A}^\dagger|A'\rangle^* = A\langle A'|A\rangle$$

However, remember that we took $\hat{A}$ to a hermitian operator, therefore $\hat{A}^\dagger = \hat{A}$, and by the second eigenvalue equation,

$$\langle A|\hat{A}|A'\rangle^* = A\langle A'|A\rangle \implies [\langle A|A'|A'\rangle]^* = A\langle A'|A\rangle$$

$$A'^*\langle A'|A\rangle = A\langle A'|A\rangle \implies (A'^* - A)\langle A'|A\rangle = 0$$

If we state A and A' to be distinct eigenvalues, the only this equality is true is when $\langle A'|A\rangle = 0$. Therefore, **the eigenkets of distinct eigenvalues are orthogonal for a hermitian operator**. If we state that the two eigenvalues are the same, then we get that $A = A^*$ and therefore **the eigenvalues of a hermitian operator are only real**.

We see that the eigenkets of a hermitian are orthonormal, and can therefore form a *basis*. Any state can therefore be represented as a linear combination of eigenkets $|A_n\rangle$ of an observable A .

$$|\mathcal{S}\rangle = \sum_{n=1} c_n |A_n\rangle \quad (1.23)$$

Since the basis is orthonormal, we can do the following:

$$\begin{aligned} \langle A_m | \mathcal{S} \rangle &= \langle A_m | \left[\sum_{n=1} c_n |A_n\rangle \right] = \sum_{n=1} c_n \langle A_m | A_n \rangle = \sum_{n=1} c_n \delta_{mn} = c_m \\ \therefore c_n &= \langle A_n | \mathcal{S} \rangle \end{aligned} \quad (1.24)$$

Here, the symbol δ_{mn} is the *Kronecker Delta*, which has the following property:

$$\delta_{mn} = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases} \quad (1.25)$$

Therefore, we can finally write the state as:

$$|\mathcal{S}\rangle = \sum_{n=1} |A_n\rangle \langle A_n | \mathcal{S} \rangle \quad (1.26)$$

Notice that the argument of the summation is actually an outer product acting on a ket. If we factor it out and consider the sum of outer products to be an operator, we see that it is the *identity operator* $\mathbf{1}$.

$$|\mathcal{S}\rangle = \left[\sum_{n=1} |A_n\rangle \langle A_n| \right] |\mathcal{S}\rangle = \mathbf{1} |\mathcal{S}\rangle$$

Identity Operator

$$\hat{I} = \sum_{n=1} |A_n\rangle \langle A_n| = \sum_{n=1} \Lambda_{A_n} \quad (1.27)$$

This result is an incredibly powerful tool. Consider an operator $\hat{Q}$ in an N dimensional space. Using our new relation,

$$\hat{Q} = \hat{I} \hat{Q} \hat{I} = \left[\sum_{n=1} |A_n\rangle \langle A_n| \right] \hat{Q} \left[\sum_{m=1} |A_m\rangle \langle A_m| \right] = \sum_{n,m=1} \langle A_n | \hat{Q} | A_m \rangle |A_n\rangle \langle A_m| \quad (1.28)$$

Here, $\langle A_n | \hat{Q} | A_m \rangle$ is the matrix element of the n^{th} row and m^{th} column of $\hat{Q}$.

$$\hat{Q} \doteq \begin{pmatrix} \langle A_1 | \hat{Q} | A_1 \rangle & \langle A_1 | \hat{Q} | A_2 \rangle & \dots & \langle A_1 | \hat{Q} | A_N \rangle \\ \langle A_2 | \hat{Q} | A_1 \rangle & \langle A_2 | \hat{Q} | A_2 \rangle & \dots & \langle A_2 | \hat{Q} | A_N \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle A_N | \hat{Q} | A_1 \rangle & \langle A_N | \hat{Q} | A_2 \rangle & \dots & \langle A_N | \hat{Q} | A_N \rangle \end{pmatrix} \quad (1.29)$$

If we consider the matrix representation of observable A in the basis of its eigenkets, we get a diagonal matrix with diagonal elements being the eigenvalues.

$$\hat{A} = \sum_{n,m=1} |A_n\rangle \langle A_n| \hat{A} |A_m\rangle \langle A_m| = \sum_{n,m=1} A_n |A_n\rangle \underbrace{\langle A_n|A_m\rangle}_{\delta_{nm}} \langle A_m| = \sum_{n=1} A_n \Lambda_{A_n} \quad (1.30)$$

1.2 Measurement and Uncertainty

1.2.1 Measurement

Let us come back to the concept of a state. We know that it is a ket and can be written as a superposition of eigenkets of an observable A . If we measure the observable A and we record that we obtain the value of A is A_1 , then it is said that *the state ket is thrown into the eigenket of the measured variable*. Before the measurement, the state ket is in a superposition of eigenkets of observable A .

$$|\mathcal{S}\rangle = \sum_{n=1} |A_n\rangle \langle A_n|\mathcal{S}\rangle \quad (1.31)$$

When the measurement is performed, the system is thrown into one of the eigenkets, say $|A_1\rangle$ of observable A .

$$|\mathcal{S}\rangle \xrightarrow{\text{measurement of } A} |A_1\rangle \quad (1.32)$$

If one measures the observable A again, then one will only obtain the same eigenvalue A_1 again. Thus, the eigenket $|A_1\rangle$ is called a *determinate state of A_1* .

Postulate 1.3: Born Interpretation

If a state ket $|\mathcal{S}\rangle$ is in a superposition of determinate states of an observable A , then the probability of measurement of the eigenvalue A_n is given by:

$$P(A_n) = |\langle A_n|\mathcal{S}\rangle|^2 \quad (1.33)$$

And the sum of the probabilities of all the eigenvalues of A is unity, and is called the normalization condition.

$$\sum_{n=1} P(A_n) = \sum_{n=1} |\langle A_n|\mathcal{S}\rangle|^2 = 1 \quad (1.34)$$

This is for a singular system, but if we were to empirically measure the probability, we must conduct a great number of measurements of an identical system of the state ket $|\mathcal{S}\rangle$, a *pure ensemble*. Suppose there are two or more linearly independent eigenkets of an observable that share the same eigenvalue. Then the eigenkets are said to *degenerate states* of A . This causes a few problems, a measurement A_n of A can no longer distinguish between eigenkets, and it is not clear to what eigenket the state collapses to. To solve this problem, we must instead use an orthonormal basis $\{|\phi_{n,k}\rangle\}$ that spans the degenerate subspace, typically of an observable B .

$$|\mathcal{S}\rangle = \frac{1}{\sqrt{A_n}} \sum_{k=1} |\phi_{n,k}\rangle \langle \phi_{n,k}|\mathcal{S}\rangle \quad (1.35)$$

Let us now talk about the expectation value.

Expectation Value

In statistics, the expectation value or the mean is defined as follows:

$$\langle A \rangle := \sum_{n=1} A_n P(A_n) \quad (1.36)$$

From equation (1.33) we can substitute the postulate,

$$\langle A \rangle = \sum_{n=1} A_n |\langle A_n | \mathcal{S} \rangle|^2 = \sum_{n=1} \langle \mathcal{S} | \underbrace{A_n | A_n \rangle}_{\hat{A} | A_n \rangle} \langle A_n | \mathcal{S} \rangle = \sum_{n=1} \langle \mathcal{S} | \hat{A} | A_n \rangle \langle A_n | \mathcal{S} \rangle \quad (1.37)$$

$$\langle A \rangle = \langle \mathcal{S} | \hat{A} \left[\underbrace{\sum_{n=1} | A_n \rangle \langle A_n |}_{\hat{I}} \right] | \mathcal{S} \rangle = \langle \mathcal{S} | \hat{A} | \mathcal{S} \rangle \quad (1.38)$$

1.2.2 Compatibility of Observables

Two observables A and B are said to be *compatible* when they commute, i.e.

$$[\hat{A}, \hat{B}] = 0 \iff \hat{A}\hat{B} = \hat{B}\hat{A} \quad (1.39)$$

They are *incompatible* if they do not commute, i.e.

$$[\hat{A}, \hat{B}] \neq 0 \quad (1.40)$$

Let us first consider the case of compatible observables in the space spanned by the eigenkets of A . If the eigenvalues of A are non-degenerate, then we can prove that the matrix elements of $\hat{B}$ in the basis of eigenkets of A are diagonal.

$$\langle A_n | [\hat{A}, \hat{B}] | A_m \rangle = \langle A_n | \hat{A}\hat{B} | A_m \rangle - \langle A_n | \hat{B}\hat{A} | A_m \rangle = (A_n - A_m) \langle A_n | \hat{B} | A_m \rangle = 0 \quad (1.41)$$

$\langle A_n | \hat{B} | A_m \rangle$ will vanish unless $A_n = A_m$, hence we can write the matrix elements of $\hat{B}$ as follows:

$$\langle A_n | \hat{B} | A_m \rangle = \delta_{nm} \langle A_n | \hat{B} | A_n \rangle \quad (1.42)$$

$$\therefore \hat{B} = \sum_{n=1} | A_n \rangle \langle A_n | \hat{B} | A_n \rangle \langle A_n | \quad (1.43)$$

Suppose $\hat{B}$ acts on an eigenket of A , say A_m ,

$$\hat{B} | A_m \rangle = \left[\sum_{n=1} | A_n \rangle \langle A_n | \hat{B} | A_n \rangle \langle A_n | \right] | A_m \rangle = \left[\langle A_m | \hat{B} | A_m \rangle \right] | A_m \rangle \quad (1.44)$$

Therefore we see that A and B share eigenkets but with different eigenvalues. These are called *simultaneous eigenkets* and are represented by $| A_n, B_n \rangle$. This holds true for degenerate states.

A simultaneous eigenket $|A_n, B_n\rangle$ has the following property:

$$\hat{A}|A_n, B_n\rangle = A_n|A_n, B_n\rangle \quad (1.45)$$

$$\hat{B}|A_n, B_n\rangle = B_n|A_n, B_n\rangle \quad (1.46)$$

If we have a set of compatible observables $A, B, C, \dots$, then the simultaneous eigenkets of $A_n, B_n, C_n, \dots$ is uniquely specified and denoted by a collective index $K_n = (A_n, B_n, C_n, \dots)$.

$$|K_n\rangle = |A_n, B_n, C_n, \dots\rangle \implies \langle K_n | K_m \rangle = \delta_{nm} \implies \sum_{n=1} \Lambda_{K_n} = \hat{I} \quad (1.47)$$

1.2.3 Incompatibility of Observables

Two observables are said to be *incompatible* if they do not commute, i.e.

$$[\hat{A}, \hat{B}] \neq 0 \iff \hat{A}\hat{B} \neq \hat{B}\hat{A} \quad (1.48)$$

This implies that they do not have simultaneous eigenkets $|A_n, B_n\rangle$. However, there can exist a subspace of the Hilbert ket space such that A and B enjoy simultaneous eigenkets with the following equation being true.

$$\hat{A}\hat{B}|A_n, B_n\rangle = \hat{B}\hat{A}|A_n, B_n\rangle \quad (1.49)$$

Let us discuss a measurement experiment to better understand incompatible observables.

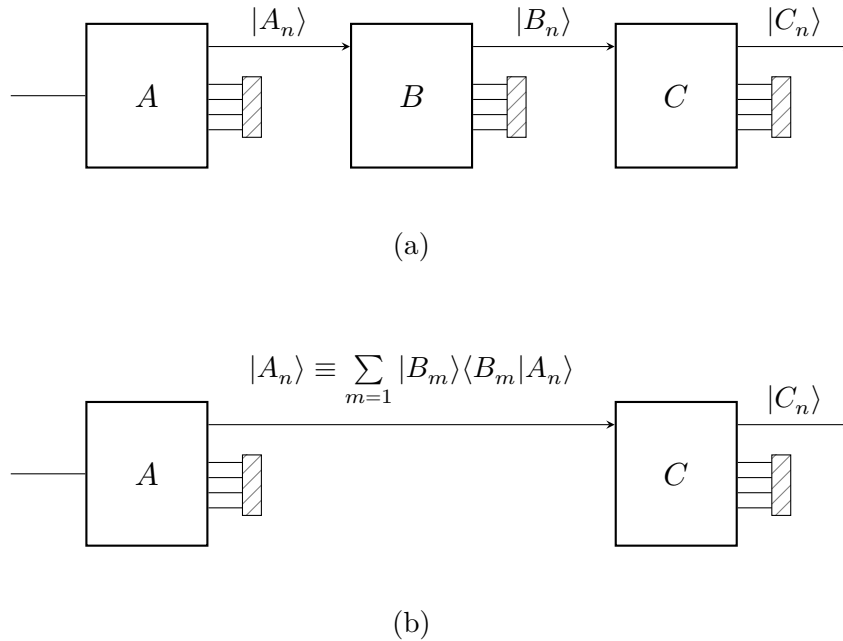


Fig 1.1: Sequential selective measurements

Consider a setup (Fig 1.1:(a)) where a state $|A_n\rangle = |A'\rangle$ is measured selectively, while all other measured states are discarded by a filter. The same is repeated for $|B_n\rangle = |B'\rangle$ and $|C_n\rangle = |C'\rangle$. The probability of a measurement of $|C'\rangle$ in the final filter would be a product of the probability that we get $|B'\rangle$ in the second filter in the eigenket basis of A and the probability that we get $|C'\rangle$ in the eigenket basis of B i.e. $|\langle C'|B'\rangle|^2 \cdot |\langle B'|A'\rangle|^2$. We repeat this procedure for all possible values of B , so we get the following sum:

$$\sum_{n=1} |\langle C'|B_n\rangle|^2 |\langle B_n|A'\rangle|^2 = \sum_{n=1} \langle C'|B_n\rangle \langle B_n|A'\rangle \langle A'|B_n\rangle \langle B_n|C'\rangle \quad (1.50)$$

If we consider a second setup (Fig 1.1(b)) where the second filter for B is inactive, the probability of measurement of C' in the final filter would be:

$$|\langle C'|A'\rangle|^2 = \left| \sum_{n=1} \langle C'|B_n\rangle \langle B_n|A'\rangle \right|^2 = \sum_{n,m=1} \langle C'|B_n\rangle \langle B_n|A'\rangle \langle A'|B_m\rangle \langle B_m|C'\rangle \quad (1.51)$$

You see that both equations are not the same! This implies that the probability indeed depends on whether a measurement of B has been taken, and that the probability are not independent of each other, unlike compatible observables. The two equations are equal if $[\hat{A}, \hat{B}] = 0$ or $[\hat{B}, \hat{C}] = 0$.

1.2.4 Uncertainty Relation

Given an observable A , we define the mean deviation operator as follows:

$$\Delta A = \hat{A} - \langle A \rangle \quad (1.52)$$

In statistics, the variance is defined as follows:

$$v = \sigma^2 = \sum_{n=1} (A_n - \langle A \rangle)^2 = \langle A^2 \rangle - \langle A \rangle^2 \quad (1.53)$$

From the new operator we just defined, we can write:

$$\sigma_A^2 = \langle (\Delta A)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2 \quad (1.54)$$

The variance, or as some books call it – dispersion – measures how much the value of A deviates from the mean value. The square root of the variance is called the uncertainty or standard deviation.

Uncertainty Principle

We consider the variance of two observables A and B .

$$\sigma_A^2 = \langle (\Delta A)^2 \rangle \quad (1.55)$$

$$\sigma_B^2 = \langle (\Delta B)^2 \rangle \quad (1.56)$$

By the Cauchy–Schwarz Inequality,

$$\sigma_A^2 \sigma_B^2 = \langle (\Delta A)^2 (\Delta B)^2 \rangle \geq |\langle \Delta A \Delta B \rangle|^2 \quad (1.57)$$

And note that we can write $\Delta A \Delta B$ as follows:

$$\Delta A \Delta B = \frac{1}{2} [\hat{A}, \hat{B}] + \frac{1}{2} \{\hat{A}, \hat{B}\} \quad (1.58)$$

$$\Rightarrow |\langle \Delta A \Delta B \rangle|^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \quad (1.59)$$

$$\therefore \sigma_A^2 \sigma_B^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \quad (1.60)$$

This is the uncertainty principle. If we consider the anti commutator term too, we get a stronger relation.

1.2.5 Unitary Operators and Change of Basis

An operator whose inverse is the same as its hermitian is called a *unitary operator*, and it follows the following identity:

$$\hat{U} \hat{U}^\dagger = \hat{U}^\dagger \hat{U} = \hat{I} \quad (1.61)$$

Now we consider an operator $\hat{U} = \sum_{n=1} |A_n\rangle \langle B_n|$ where A_n and B_n are eigenkets of observables A and B . We see that this operator is unitary.

$$\hat{U}^\dagger = \sum_{n=1} (|B_n\rangle \langle A_n|)^\dagger = \sum_{n=1} |A_n\rangle \langle B_n| \quad (1.62)$$

$$\hat{U}^\dagger \hat{U} = \sum_{n,m=1} \langle B_n | B_m \rangle \langle A_m | A_n \rangle = \sum_{n,m=1} \delta_{nm} \delta_{nm} \hat{I} = \hat{I} \quad (1.63)$$

Applying this operator to an eigenket of B , say $|B_m\rangle$:

$$\hat{U} |B_m\rangle = \sum_{n=1} |A_n\rangle \langle B_n | B_m \rangle = |A_m\rangle \quad (1.64)$$

Similarly, operating $\hat{U}^\dagger$ to $|A_m\rangle$,

$$\hat{U}^\dagger |A_m\rangle = \sum_{n=1} |B_n\rangle \langle A_n | A_m \rangle = |B_m\rangle \quad (1.65)$$

We see that the unitary operator transforms eigenkets of B to A and its hermitian, vice-versa. For some arbitrary ket $|\alpha\rangle$ in the basis of eigenkets of A , we multiply $\langle B_m|$ to the left:

$$\langle B_m|\alpha\rangle = \sum_{n=1} \langle B_m|A_n\rangle \langle A_n|\alpha\rangle = \sum_{n=1} \langle A_m|\hat{U}^\dagger|A_n\rangle \langle A_n|\alpha\rangle \quad (1.66)$$

Therefore, we get the elements of $|\alpha\rangle$ in the basis of eigenkets of B by just operating $\hat{U}^\dagger$.

$$\langle B_n|\alpha\rangle = \langle A_n|\hat{U}^\dagger|\alpha\rangle \iff |\alpha\rangle = \sum_{n=1} |B_n\rangle \langle A_n|\hat{U}^\dagger|\alpha\rangle \quad (1.67)$$

Similarly, the matrix elements for an operator are given by:

$$\langle B_m|\hat{Q}|B_n\rangle = \sum_{i,j=1} \langle B_m|A_i\rangle \langle A_i|\hat{Q}|A_j\rangle \langle A_j|B_n\rangle \quad (1.68)$$

$$= \sum_{i,j=1} \langle A_m|\hat{U}^\dagger|A_i\rangle \langle A_i|\hat{Q}|A_j\rangle \langle A_j|\hat{U}|A_n\rangle \quad (1.69)$$

$$\therefore \hat{Q}_B = \hat{U}^\dagger \hat{Q}_A \hat{U} \quad (1.70)$$

Unitary Equivalent Observables

If we instead consider an observable A , then its unitary transformation $\hat{A}_B = \hat{U}^\dagger \hat{A} \hat{U}$ has the same eigenvalues, but the eigenkets are that of the B .

$$\hat{A}|A_n\rangle = A_n|A_n\rangle \implies \hat{U} \hat{A} \hat{U}^\dagger \hat{U}|A_n\rangle = A_n \hat{U}|A_n\rangle \quad (1.71)$$

$$(\hat{U} \hat{A} \hat{U}^\dagger)|B_n\rangle = A_n|B_n\rangle \quad (1.72)$$

Then we call $\hat{A}$ and $\hat{U} \hat{A} \hat{U}^\dagger$ *unitary equivalent observables*. It means that they have identical spectra of eigenvalues. In some cases, we may see that $\hat{B}$ and $\hat{U} \hat{A} \hat{U}^\dagger$ are in fact the same.

Unitary operators can also be used diagonalize other operators. For example, if we know $\hat{B}$ in the basis of eigenkets of A , we can find the eigenvalues by diagonalizing $\hat{B}_A$.

$$\hat{B}|B_n\rangle = B_n|B_n\rangle \implies \langle A_m|\hat{B}|B_n\rangle = B_n \langle A_m|B_n\rangle \quad (1.73)$$

$$\sum_{n=1} \langle A_m|\hat{B}|A_n\rangle \langle A_n|B_n\rangle = B_n \langle A_m|B_n\rangle \quad (1.74)$$

In matrix form, we can write it as:

$$\begin{pmatrix} B_{11} & B_{12} & \dots \\ B_{21} & B_{22} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} C_{1m} \\ C_{2m} \\ \vdots \end{pmatrix} = B_m \begin{pmatrix} C_{1m} \\ C_{2m} \\ \vdots \end{pmatrix} \quad (1.75)$$

$$B_{ij} = \langle A_i|\hat{B}|A_j\rangle \quad C_{nm} = \langle A_n|B_m\rangle \quad (1.76)$$

The matrix formed by the elements C_{nm} is the unitary operator that diagonalizes $\hat{B}_A$ into $\hat{B}$.

1.3 Position and Momentum

1.3.1 Continuous Spectra

Until now, we only discussed observables with a discrete spectra. However, we have observables like position and momentum that take continuous values. To analyze these observables, we just swap the discrete sum with a continuous integral.

For some observable $\hat{e}$ with the following continuous eigenvalue equation, we generalize the properties of discrete spectra.

$$\hat{e}|e\rangle = e|e\rangle \quad (1.77)$$

$$\text{Orthonormality: } \langle A_n | A_m \rangle = \delta_{nm} \longrightarrow \langle e | e' \rangle = \delta(e - e') \quad (1.78)$$

$$\text{Identity Operator: } \sum_{n=1} |A_n\rangle \langle A_n| = \hat{I} \longrightarrow \int |e\rangle \langle e| de = \hat{I} \quad (1.79)$$

$$\text{Born Rule: } \sum_{n=1} |\langle A_n | \alpha \rangle|^2 = 1 \longrightarrow \int |\langle e | \alpha \rangle|^2 dz = 1 \quad (1.80)$$

$$\text{Ket Expansion: } |\alpha\rangle = \sum_{n=1} |A_n\rangle \langle A_n | \alpha \rangle \longrightarrow |\alpha\rangle = \int |e\rangle \langle e | \alpha \rangle de \quad (1.81)$$

$$\text{Matrix Expansion: } \langle A_m | \hat{A} | A_n \rangle = A_n \delta_{mn} \longrightarrow \langle e_{z'} | \hat{e} | e \rangle = e \delta(e - e') \quad (1.82)$$

Here the symbol $\delta(x)$ stands for the Dirac delta function, which has the two following properties:

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad (1.83)$$

$$\text{For } \epsilon > 0, \int_{-\epsilon}^{+\epsilon} \delta(x) dx = 1 \quad (1.84)$$

1.3.2 Position

Consider the eigenvalue equation for position in one dimension:

$$\hat{x}|x\rangle = x|x\rangle \quad (1.85)$$

If we take a measurement of the position, say we obtain x' , then the system collapses to the corresponding eigenket $|x'\rangle$. Of course, this is a very ideal case, real detector will record a measurement between $x' \pm \epsilon$.

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |x\rangle \langle x | \mathcal{S} \rangle dx \xrightarrow{\text{measurement of } x} \int_{x'-\epsilon}^{x'+\epsilon} |x\rangle \langle x | \mathcal{S} \rangle dx \quad (1.86)$$

And by the Born Probability Interpretation, the probability that we obtain a measurement between x_1 and x_2 is:

$$P(x_1 < x < x_2) = \int_{x_1}^{x_2} |\langle x|\mathcal{S}\rangle|^2 dx \quad (1.87)$$

Extending this into three dimensions, we can write the state ket as follows:

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r \quad (1.88)$$

where $|\mathbf{r}\rangle$ is the simultaneous eigenket of observables x, y and z .

$$|\mathbf{r}\rangle = |x, y, z\rangle \quad (1.89)$$

This implies that the observables x, y, z are all compatible with each other.

1.3.3 Translation

We introduce an operator that changes a position eigenket $|\mathbf{r}\rangle$ into $|\mathbf{r} + d\mathbf{r}\rangle$. We can say the ket is being translated by an infinitesimal amount $d\mathbf{r}$ by the *translation operator*.

$$\mathcal{T}(d\mathbf{r})|\mathbf{r}\rangle = |\mathbf{r} + d\mathbf{r}\rangle \quad (1.90)$$

Let us further analyze this by translating an arbitrary state ket.

$$\mathcal{T}(d\mathbf{r})|\mathcal{S}\rangle = \mathcal{T}(d\mathbf{r}) \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r = \int_{-\infty}^{+\infty} |\mathbf{r} + d\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r} - d\mathbf{r}|\mathcal{S}\rangle d^3r \quad (1.91)$$

The last equality is true because we are integrating over all space, and the substitution $\mathbf{r} \rightarrow \mathbf{r} - d\mathbf{r}$ will not change the integral.

We demand that the translation operator must have the following properties:

- The operator must be unitary by the fact that the probability must be conserved upon translation.

$$\langle \mathcal{S}|\mathcal{S}\rangle = \langle \mathcal{S}|\mathcal{T}^\dagger(d\mathbf{r})\mathcal{T}(d\mathbf{r})|\mathcal{S}\rangle \iff \mathcal{T}^\dagger(d\mathbf{r})\mathcal{T}(d\mathbf{r}) = \hat{I} \quad (1.92)$$

- The net result of two translations by $d\mathbf{r}$ and $d\mathbf{r}'$ should be their sum.

$$\mathcal{T}(d\mathbf{r})\mathcal{T}(d\mathbf{r}') = \mathcal{T}(d\mathbf{r} + d\mathbf{r}') \quad (1.93)$$

- The inverse of a translation should be the same as the translation in the opposite direction.

$$\mathcal{T}(-d\mathbf{r}) = \mathcal{T}^{-1}(d\mathbf{r}) = \mathcal{T}^\dagger(d\mathbf{r}) \quad (1.94)$$

- The translation such that $d\mathbf{r} \rightarrow 0$ must be an identity.

$$\lim_{d\mathbf{r} \rightarrow 0} \mathcal{T}(d\mathbf{r}) = \hat{I} \quad (1.95)$$

All these properties are satisfied if we demand that the translation operator and the identity have a difference in the first order, i.e. $(\hat{\mathbf{K}} = (\hat{K}_x, \hat{K}_y, \hat{K}_z))$ is a hermitian operator)

$$\mathcal{T}(d\mathbf{r}) = \hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r} \quad (1.96)$$

Checking the first property:

$$\mathcal{T}(d\mathbf{r})\mathcal{T}^\dagger(d\mathbf{r}) = (\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})(\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})^\dagger \quad (1.97)$$

$$= (\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})(\hat{I} + i\hat{\mathbf{K}} \cdot d\mathbf{r}) \quad (1.98)$$

$$= \hat{I} + i(0) \cdot d\mathbf{r} + \hat{\mathbf{K}}^2 [d\mathbf{r}]^2 \approx \hat{I} \quad (1.99)$$

Checking the second property:

$$\mathcal{T}(d\mathbf{r})\mathcal{T}(d\mathbf{r}') = (\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})(\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r}') \quad (1.100)$$

$$= \hat{I} - i\hat{\mathbf{K}}(d\mathbf{r} + d\mathbf{r}') + \hat{\mathbf{K}}^2 \cdot d\mathbf{r} \cdot d\mathbf{r}' \quad (1.101)$$

$$\approx \hat{I} - i\hat{\mathbf{K}}(d\mathbf{r} + d\mathbf{r}') = \mathcal{T}(d\mathbf{r} + d\mathbf{r}') \quad (1.102)$$

And the third and fourth properties are also satisfied. Now, consider the following:

$$\hat{\mathbf{r}}\mathcal{T}(d\mathbf{r})|\mathbf{r}\rangle = \hat{\mathbf{r}}|\mathbf{r} + d\mathbf{r}\rangle = (\mathbf{r} + d\mathbf{r})|\mathbf{r} + d\mathbf{r}\rangle \quad (1.103)$$

$$\mathcal{T}(d\mathbf{r})\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}\mathcal{T}(d\mathbf{r})|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r} + d\mathbf{r}\rangle \quad (1.104)$$

From these two, we see that:

$$[\hat{\mathbf{r}}, \mathcal{T}(d\mathbf{r})]|\mathbf{r}\rangle = d\mathbf{r}|\mathbf{r} + d\mathbf{r}\rangle \approx d\mathbf{r}|\mathbf{r}\rangle \quad (1.105)$$

$$\therefore [\hat{\mathbf{r}}, \mathcal{T}(d\mathbf{r})] = d\mathbf{r} \quad (1.106)$$

We can generate a relation between $\hat{\mathbf{r}}$ and $\hat{\mathbf{K}}$ using this relation.

$$-i\hat{\mathbf{r}}\hat{\mathbf{K}} \cdot d\mathbf{r} + i\hat{\mathbf{K}} \cdot d\mathbf{r} \cdot \hat{\mathbf{r}} = d\mathbf{r} \quad (1.107)$$

If we check for $\hat{\mathbf{r}}$ and $\hat{\mathbf{K}}$ in each direction, we get the following relation:

$$[\hat{r}_i, \hat{K}_j] = i\delta_{ij} \quad (1.108)$$

where $i, j = x, y, z$ and $\hat{r}_x = \hat{x}$, $\hat{r}_y = \hat{y}$ and $\hat{r}_z = \hat{z}$.

1.3.4 Momentum

If we realize that momentum, classically, can be thought of a generator of translation. It just so happens that $\hat{\mathbf{K}}$ term in the translation operator is the wave number, and by the De-Broglie relation,

$$k = \frac{p}{\hbar} \implies \hat{\mathbf{K}} = \frac{1}{\hbar} \hat{\mathbf{p}} \quad (1.109)$$

Therefore, the translation operator can be written as follows:

$$\mathcal{T}(d\mathbf{r}) = \hat{I} - i\hat{\mathbf{p}} \cdot d\mathbf{r}/\hbar \quad (1.110)$$

And we obtain the following commutator relation:

Canonical Commutation Relation

$$[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij} \quad (1.111)$$

This is called canonical commutation relation of quantum mechanics. From the uncertainty principle, we can now obtain the famous Heisenberg Uncertainty Principle between position and momentum.

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2} \quad (1.112)$$

Consider a finite translation Δx in the $\mathbf{x}$ direction.

$$\mathcal{T}(\Delta x \cdot \mathbf{x})|x\rangle = |\mathbf{r} + \Delta x \cdot \mathbf{x}\rangle \quad (1.113)$$

We can achieve this N by repeated applications of the translation operator by some $d\mathbf{r} = \Delta x \cdot \mathbf{x}/N$ and letting $N \rightarrow \infty$.

$$\mathcal{T}(\Delta x \cdot \mathbf{x}) = \lim_{N \rightarrow \infty} \mathcal{T}^N \left(\frac{\Delta x}{N} \mathbf{x} \right) = \lim_{N \rightarrow \infty} \left(1 - \frac{i\hat{p}_x \Delta x}{N\hbar} \right)^N = e^{i\hat{p}_x \Delta x/\hbar} \quad (1.114)$$

By the use of a Taylor series expansion,

$$\mathcal{T}(\Delta x \cdot \mathbf{x}) = 1 - \frac{i\hat{p}_x \Delta x}{\hbar} - \frac{\hat{p}_x^2 (\Delta x)^2}{2\hbar^2} + \dots \quad (1.115)$$

Similarly, in the $\mathbf{y}$ direction we have,

$$\mathcal{T}(\Delta y \cdot \mathbf{y}) = 1 - \frac{i\hat{p}_y \Delta y}{\hbar} - \frac{\hat{p}_y^2 (\Delta y)^2}{2\hbar^2} + \dots \quad (1.116)$$

Now, we consider the commutator of these two operators

$$[\mathcal{T}(\Delta x \cdot \mathbf{x}), \mathcal{T}(\Delta y \cdot \mathbf{y})] = \mathcal{T}(\Delta x \cdot \mathbf{x})\mathcal{T}(\Delta y \cdot \mathbf{y}) - \mathcal{T}(\Delta y \cdot \mathbf{y})\mathcal{T}(\Delta x \cdot \mathbf{x}) \quad (1.117)$$

$$= \mathcal{T}(\Delta x \cdot \mathbf{x} + \Delta y \cdot \mathbf{y}) - \mathcal{T}(\Delta y \cdot \mathbf{y} + \Delta x \cdot \mathbf{x}) = 0 \quad (1.118)$$

And if we find the commutator in the series expansion form, we obtain that:

$$\frac{(\Delta x)(\Delta y)}{\hbar^2} [\hat{p}_x, \hat{p}_y] = 0 \implies [\hat{p}_x, \hat{p}_y] = 0 \quad (1.119)$$

And more generally we have:

$$[\hat{p}_i, \hat{p}_j] = 0 \quad (1.120)$$

Therefore, we have a simultaneous eigenket $|\mathbf{p}\rangle$ of p_x, p_y and p_z .

$$|\mathbf{p}\rangle = |p_x, p_y, p_z\rangle \quad (1.121)$$

And if we work out the effect of the translation operator on $|\mathbf{p}\rangle$,

$$\mathcal{T}(d\mathbf{r})|\mathbf{p}\rangle = \left(1 - \frac{i\hat{\mathbf{p}} \cdot d\mathbf{r}}{\hbar}\right) |\mathbf{p}\rangle = \left(1 - \frac{i\mathbf{p} \cdot d\mathbf{r}}{\hbar}\right) |\mathbf{p}\rangle \quad (1.122)$$

$|\mathbf{p}\rangle$ is therefore an eigenket of the translation operator, and the operators commute.

$$[\mathcal{T}(d\mathbf{r}), \hat{\mathbf{p}}] = 0 \quad (1.123)$$

1.3.5 Wave Functions

We finally arrive at a study of wave functions in the position and momentum basis. We know that any state is normalized as $\langle x'|x\rangle = \delta(x' - x)$, which leads to the following ket expansion:

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.124)$$

and we can treat $|\langle x|\mathcal{S}\rangle|^2 dx$ to be the probability of measuring x in an infinitesimal interval of dx . We interpret that inner product $\langle x|\mathcal{S}\rangle$ is the position space wave function $\psi(x)$.

$$\psi_{\mathcal{S}}(x) = \langle x|\mathcal{S}\rangle \quad (1.125)$$

The inner product between two arbitrary vectors $|\alpha\rangle$ and $|\beta\rangle$ can be written as:

$$\langle \beta|\alpha\rangle = \int_{-\infty}^{+\infty} \langle \beta|x\rangle \langle x|\alpha\rangle dx = \int_{-\infty}^{+\infty} \psi_{\beta}^*(x) \psi_{\alpha}(x) dx \quad (1.126)$$

If we have some state as a superposition of eigenkets of observable A , we can write the following:

$$|\mathcal{S}\rangle = \sum_{n=1} |A_n\rangle \langle A_n|\mathcal{S}\rangle \implies \psi_{\mathcal{S}}(x) = \sum_{n=1} c_n u_n(x) \quad (1.127)$$

where $c_n = \langle A_n|\mathcal{S}\rangle$ and $u_n(x) = \langle x|A_n\rangle$ is the eigenfunction of operator $\hat{A}$ with eigenvalue A_n .

$$\hat{A}u_n(x) = A_n u_n(x) \quad (1.128)$$

The expectation value of an observable A can be written as follows:

$$\langle A\rangle = \langle \mathcal{S}|\hat{A}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \langle \mathcal{S}|x\rangle \langle x|\hat{A}|x'\rangle \langle x'|\mathcal{S}\rangle dx dx' \quad (1.129)$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi_{\mathcal{S}}^*(x) \langle x|\hat{A}|x'\rangle \psi_{\mathcal{S}}(x') dx dx' \quad (1.130)$$

If $\hat{A} = g(x)$, then we can do the following:

$$\langle A \rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi_S^*(x) g(x) \underbrace{\langle x|x' \rangle}_{\delta(x-x')} \psi_S(x) dx dx' \quad (1.131)$$

$$= \int_{-\infty}^{+\infty} \psi_S^*(x) g(x) \psi_S(x) dx \quad (1.132)$$

We can say the same about the wave function in momentum space.

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |p\rangle \langle p|\mathcal{S}\rangle dp \quad (1.133)$$

$$\phi_S(p) = \langle p|\mathcal{S}\rangle = \sum_{n=1} c_n v_n(p) \quad (1.134)$$

$$\text{where } c_n = \langle p|A_n\rangle, \hat{A}v_n(p) = A_nv_n(p) \quad (1.135)$$

$$\langle A \rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \phi_S^*(p) \langle p|\hat{A}|p'\rangle \phi_S(p') dp dp' = \int_{-\infty}^{+\infty} \phi_S^*(p) \underbrace{h(p)}_{\hat{A}=h(p)} \phi_S(p) dp \quad (1.136)$$

1.3.6 Momentum and Position Operators

First, we may examine how the momentum operators look in the position basis. Consider the definition of momentum as a generator of translations, hence:

$$\left(1 - \frac{i\hat{p}\Delta x}{\hbar}\right) |\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \mathcal{T}(\Delta x) |x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.137)$$

$$= \int_{-\infty}^{+\infty} |x + \Delta x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.138)$$

$$= \int_{-\infty}^{+\infty} |x\rangle \langle x - \Delta x|\mathcal{S}\rangle dx \quad (1.139)$$

$$= \int_{-\infty}^{+\infty} |x\rangle (\langle x|\mathcal{S}\rangle - \Delta x \partial_x \langle x|\mathcal{S}\rangle) dx \quad (1.140)$$

We get the last line by applying a Taylor series expanding and ignoring higher order terms. Now, by comparing the left and right hand sides, we obtain:

$$\hat{p}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} (-i\hbar \partial_x \langle x|\mathcal{S}\rangle) |x\rangle dx \quad (1.141)$$

By orthogonality, it implies the following:

$$\langle x|\hat{p}|\mathcal{S}\rangle = -i\hbar \partial_x \langle x|\mathcal{S}\rangle \quad (1.142)$$

And therefore, the momentum operator in the position basis is given by:

$$\therefore \langle x|\hat{p} = -i\hbar \partial_x \langle x| \quad (1.143)$$

Since we have the momentum operator purely as function of x , its expectation value is given by:

$$\langle p \rangle = \int_{-\infty}^{+\infty} \psi^*(x) [-i\hbar \partial_x \psi(x)] dx \quad (1.144)$$

Consider the momentum eigenvalue equation:

$$\hat{p}|p\rangle = p|p\rangle \implies \langle x|\hat{p}|p\rangle = p\langle x|p\rangle \quad (1.145)$$

And from earlier, we know that $\langle x|\hat{p}|\mathcal{S}\rangle = -i\hbar \partial_x \langle x|\mathcal{S}\rangle$, then we get the following differential equation:

$$\partial_x \langle x|p\rangle = \frac{ip}{\hbar} \langle x|p\rangle \quad (1.146)$$

The solution to this differential equation is $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$ after normalization.

Now consider the following:

$$\phi(p) = \langle p|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \langle p|x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.147)$$

$$\therefore \phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-ipx/\hbar} \psi(x) dx \quad (1.148)$$

Similarly,

$$\therefore \psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{ipx/\hbar} \phi(p) dp \quad (1.149)$$

This way, we transform between the position and momentum wave function, and therefore the function $\langle x|p\rangle$ is called the transformation function of x and p .

Now, to find the position operator in momentum space, consider the following:

$$\langle p|\hat{x}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \langle p|\hat{x}|x\rangle \langle x|\mathcal{S}\rangle dx = \int_{-\infty}^{+\infty} x \langle p|x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.150)$$

Since we know that the form of $\langle x|p\rangle$ and that $\langle x|\mathcal{S}\rangle = \psi(x)$,

$$\langle p|\hat{x}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} x e^{-ipx/\hbar} \psi(x) dx = \int_{-\infty}^{+\infty} i\hbar \partial_p (e^{-ipx/\hbar}) \psi(x) dx \quad (1.151)$$

We can pull this derivative out as it is in terms of p , not x , which gives us:

$$\langle p|\hat{x}|\mathcal{S}\rangle = i\hbar \partial_p \underbrace{\int_{-\infty}^{+\infty} e^{-ipx/\hbar} \psi(x) dx}_{\phi(p)} = i\hbar \partial_p \langle p|\mathcal{S}\rangle \quad (1.152)$$

And therefore, we obtain the position operator in the momentum basis to be the following:

$$\therefore \langle p|\hat{x} = i\hbar \partial_p \langle p| \quad (1.153)$$

1.3.7 Generalization to Three Dimensions

The eigenkets of position and momentum satisfy:

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle \quad \hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle \quad (1.154)$$

They obey the normalization conditions

$$\langle \mathbf{r}|\mathbf{r}'\rangle = \delta^3(\mathbf{r} - \mathbf{r}') \quad \langle \mathbf{p}|\mathbf{p}'\rangle = \delta^3(\mathbf{p} - \mathbf{p}') \quad (1.155)$$

The identity operator is given by:

$$\hat{I} = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}| d^3r = \int_{-\infty}^{+\infty} |\mathbf{p}\rangle \langle \mathbf{p}| d^3p \quad (1.156)$$

The expansion of a ket is given by:

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r = \int_{-\infty}^{+\infty} |\mathbf{p}\rangle \langle \mathbf{p}|\mathcal{S}\rangle d^3p \quad (1.157)$$

where $\psi(\mathbf{r}) = \langle \mathbf{r}|\mathcal{S}\rangle$ and $\phi(\mathbf{p}) = \langle \mathbf{p}|\mathcal{S}\rangle$ are the position and momentum space wave functions respectively.

The operators in each other basis given by:

$$\langle \mathbf{r}|\hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{r}}^2 \langle \mathbf{r}| \quad \langle \mathbf{p}|\hat{\mathbf{r}} = i\hbar \nabla_{\mathbf{p}}^2 \langle \mathbf{p}| \quad (1.158)$$

and their transformation function is therefore:

$$\langle \mathbf{r}|\mathbf{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \quad (1.159)$$

1.4 Schrödinger Equation

1.4.1 Time Evolution

We consider a state represented by the state ket $|\mathcal{S}(t_0)\rangle$ at some time t_0 . At some later time $t > t_0$, we denote the state by $|\mathcal{S}, t_0; t\rangle$. At $t \rightarrow t_0$, we just write $|\mathcal{S}, t_0\rangle$. We wish to analyze the time evolution of this state, and hence we define the *time-evolution operator*.

$$\mathcal{U}(t, t_0)|\mathcal{S}, t_0\rangle = |\mathcal{S}, t_0; t\rangle \quad (1.160)$$

We demand the following properties from the time-evolution operator.

- Probability must be conserved, and hence it must be unitary.

$$\mathcal{U}(t, t_0)\mathcal{U}^\dagger(t, t_0) = \hat{I} \quad (1.161)$$

- The composition property, which states that a time evolution of from t_0 to t_2 can be

achieved using two evolutions, one from t_0 to t_1 and from t_1 to t_2 , where $t_0 < t_1 < t_2$.

$$\mathcal{U}(t_2, t_0) = \mathcal{U}(t_2, t_1)\mathcal{U}(t_1, t_0) \quad (1.162)$$

We also consider an infinitesimal time-evolution operator, similar to our translation operator for position.

$$\mathcal{U}(t_0 + dt, t_0)|\mathcal{S}, t_0\rangle = |\mathcal{S}, t_0 + dt\rangle \quad (1.163)$$

And it also has properties similar to the infinitesimal translation operator.

$$\lim_{dt \rightarrow 0} \mathcal{U}(t_0 + dt, t_0) = \hat{I} \quad (1.164)$$

$$\mathcal{U}(t_0 + dt, t_0) = 1 - i\hat{\Omega} \cdot dt \quad (1.165)$$

where $\hat{\Omega}$ is a hermitian operator with dimensions of frequency, and this form satisfies all the properties we've set for the operator. From the Planck-Einstein relation, we know that $E = \hbar\omega$ and the energy is the same as the Hamiltonian, so we write the following:

$$\hat{\Omega} = \frac{\hat{H}}{\hbar} \quad (1.166)$$

and therefore the infinitesimal time-evolution operator is as follows:

$$\mathcal{U}(t_0 + dt, t_0) = 1 - \frac{i\hat{H}}{\hbar}dt \quad (1.167)$$

1.4.2 Schrödinger Equation

Postulate 1.4: Schrödinger Equation

First, we use the composition property of the time-evolution operator.

$$\mathcal{U}(t + dt, t_0) = \mathcal{U}(t + dt, t)\mathcal{U}(t, t_0) = \left(1 - \frac{i\hat{H}}{\hbar}dt\right)\mathcal{U}(t, t_0) \quad (1.168)$$

$$\frac{\mathcal{U}(t + dt, t_0) - \mathcal{U}(t, t_0)}{dt} = -\frac{i}{\hbar}\hat{H}\mathcal{U}(t, t_0) \quad (1.169)$$

We identify that the left hand side can be written as a partial derivative with respect to time.

$$i\hbar\partial_t\mathcal{U}(t, t_0) = \hat{H}\mathcal{U}(t, t_0) \quad (1.170)$$

This is the Schrödinger Equation in terms of the time-evolution operator, and for a state ket $|\mathcal{S}, t_0\rangle$, we obtain the following:

$$i\hbar\partial_t\mathcal{U}(t, t_0)|\mathcal{S}, t_0\rangle = \hat{H}\mathcal{U}(t, t_0)|\mathcal{S}, t_0\rangle \quad (1.171)$$

$$\therefore i\hbar\partial_t|\mathcal{S}, t_0; t\rangle = \hat{H}|\mathcal{S}, t_0; t\rangle \quad (1.172)$$

The Schrödinger Equation is a *postulate*, and any derivation of it must involve an assumption, here we assume the hermitian operator $\hat{\Omega} = \hat{H}/\hbar$ in the infinitesimal time-evolution operator. Let us first find solutions to the Schrödinger Equation for the time evolution operator. We consider three cases.

Case 1 — Hamiltonian is time independent, then the time evolution operator is given by:

$$\mathcal{U}(t, t_0) = \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] \quad (1.173)$$

This can be derived by performing N infinitesimal operations with $N \rightarrow \infty$ to obtain a finite time evolution.

Case 2 — Hamiltonian is time-dependent, but commutes at different times. The solution in this case is:

$$\mathcal{U}(t, t_0) = \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t) dt \right] \quad (1.174)$$

Case 3 — Hamiltonian is time-dependent and does *not commute* at different times. The formal solution in such a case is the *Dyson Series*:

$$\mathcal{U}(t, t_0) = 1 + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \int_{t_0}^{t_2} dt_3 \dots \int_{t_0}^{t_{n-1}} \hat{H}(t_1) \hat{H}(t_2) \dots \hat{H}(t_n) dt_n \quad (1.175)$$

1.4.3 Energy Eigenkets

We consider an observable that is compatible with the Hamiltonian, which then have simultaneous eigenkets called energy eigenkets, for which we reserve the symbol $|n\rangle$.

$$\hat{H}|n\rangle = E_n|n\rangle \quad (1.176)$$

We consider an time independent Hamiltonian, and then we can expand the time-evolution operator as follows:

$$\exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] = \sum_{n,m=1} |n\rangle \langle n| \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] |m\rangle \langle m| \quad (1.177)$$

$$= \sum_{n=1} |n\rangle \langle n| \exp \left[-\frac{i}{\hbar} E_n(t - t_0) \right] \quad (1.178)$$

And therefore the time evolution of a ket would be:

$$|\mathcal{S}, t_0; t\rangle = \mathcal{U}(t, t_0) |\mathcal{S}, t_0\rangle = \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] |\mathcal{S}, t_0\rangle \quad (1.179)$$

$$= \sum_{n=1} |n\rangle \langle n|\mathcal{S}\rangle \exp \left[-\frac{i}{\hbar} E_n(t - t_0) \right] \quad (1.180)$$

We therefore see that the coefficients evolve with time by a relative phase change, while their magnitude is constant. Now we consider the expectation value of an observable B , that need not

be compatible with H .

$$\langle B \rangle = \sum_{n=1} \langle n | \mathcal{U}^\dagger(t, t_0) \hat{B} \mathcal{U}(t, t_0) | n \rangle \quad (1.181)$$

$$= \sum_{n=1} \langle n | \exp \left[\frac{i}{\hbar} \hat{H}(t - t_0) \right] \hat{B} \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] | n \rangle \quad (1.182)$$

$$= \sum_{n=1} \langle n | \hat{B} | n \rangle = \langle \mathcal{S} | \hat{B} | \mathcal{S} \rangle \quad (1.183)$$

We see that the expectation value of the observable B is independent of time, and for this reason the energy eigenket is also referred to as a *stationary state*. Similarly, for a non-stationary state, we get the expectation value to be:

$$\langle B \rangle = \sum_{n,m=1} c_n^* c_m \langle A_n | \hat{B} | A_m \rangle \exp \left[-\frac{i}{\hbar} (E_m - E_n)(t - t_0) \right] \quad (1.184)$$

1.4.4 Schrödinger's Wave Equation

We consider the position basis wave function, now with an additional time dependent factor.

$$\Psi(\mathbf{r}, t) = \langle \mathbf{r} | \mathcal{S}, t_0; t \rangle \quad (1.185)$$

The Hamiltonian operator is taken to be:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{r}, t) \quad (1.186)$$

Now, from Schrödinger's Equation we write the following:

$$i\hbar \partial_t \langle \mathcal{S}, t_0; t \rangle = \hat{H} \langle \mathcal{S}, t_0; t \rangle \implies i\hbar \partial_t \langle \mathbf{r} | \mathcal{S}, t_0; t \rangle = \langle \mathbf{r} | \hat{H} | \mathcal{S}, t_0; t \rangle \quad (1.187)$$

We know that $\langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla \langle \mathbf{r} |$, and we write the following wave equation:

$$i\hbar \partial_t \Psi(\mathbf{r}, t) = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\mathbf{r}, t) + V(\mathbf{r}, t) \Psi(\mathbf{r}, t) \quad (1.188)$$

This is fundamental equation of what is known as *wave mechanics*, called the Time-Dependent Schrödinger Wave Equation. We can further reduce this it by considering a time-independent Hamiltonian, which allows us to use:

$$\langle \mathbf{r} | \mathcal{S}, t_0; t \rangle = \langle \mathbf{r} | \mathcal{S}, t_0 \rangle e^{-iE_n t/\hbar} \iff \Psi(\mathbf{r}, t) = \psi(\mathbf{r}) e^{-iE_n t/\hbar} \quad (1.189)$$

Substitue this into the Time-Dependent Wave equation, and we obtain the following equation:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \psi(\mathbf{r}) = E \psi(\mathbf{r}) \quad (1.190)$$

This is the Time-Independent Schrödinger Wave Equation, which laid the ground work for Quantum Mechanics.

1.4.5 Interpretation of the Wave Function

The wave function is regarded to be *probability density*, which follows from the fact that it is the expansion coefficient of a state ket that determine the probability of the state ket collapsing to the corresponding eigenket.

$$\rho(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2 \quad (1.191)$$

From the time-dependent wave equation, we can do the following:

$$i\hbar\Psi^*(\mathbf{r}, t)\partial_t\Psi(\mathbf{r}, t) = -\frac{\hbar^2}{2m}\Psi^*(\mathbf{r}, t)\nabla^2\Psi(\mathbf{r}, t) + V(\mathbf{r}, t)|\Psi(\mathbf{r}, t)|^2 \quad (1.192)$$

$$i\hbar\Psi(\mathbf{r}, t)\partial_t\Psi^*(\mathbf{r}, t) = \frac{\hbar^2}{2m}\Psi(\mathbf{r}, t)\nabla^2\Psi^*(\mathbf{r}, t) - V(\mathbf{r}, t)|\Psi(\mathbf{r}, t)|^2 \quad (1.193)$$

Adding the two equations (1.192) and (1.193), we get:

$$\partial_t \underbrace{(\Psi^*(\mathbf{r}, t)\Psi(\mathbf{r}, t))}_{|\Psi(\mathbf{r}, t)|^2} = -\nabla \cdot \left(\frac{i\hbar}{2m}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) \right) \quad (1.194)$$

If we define the *probability flux* as following:

$$\mathbf{j}(\mathbf{r}, t) = \frac{i\hbar}{2m}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) \quad (1.195)$$

We obtain the continuity equation for the probability flow.

$$\partial_t\rho + \nabla \cdot \mathbf{j} = 0 \quad (1.196)$$

We also find a relation between the momentum and the probability flux, which are related by the following equation:

$$\langle p \rangle = m \int_{-\infty}^{+\infty} \mathbf{j}(\mathbf{r}, t) d^3r \quad (1.197)$$

2 ONE DIMENSIONAL PROBLEMS

We start by an analysis of several one dimensional, time-independent potentials. We will have a discussion of discrete and continuous energy spectrums, which is dependent on the potential analyzed, and a subsection on scattering experiments. Lastly, we introduce the harmonic oscillator potential and its quirks.

2.0.1 Bound and Scattering States

In our discussion of various potentials, we will encounter two types of solutions to the Schrödinger Equation — those with discretized energy states ; those with continuous energy spectra. These are called *bound states* and *scattering states* respectively.

The bound state solution is characterized by discrete spectrum of energy, and physically realisable solution. The energies of a bound state solution must be less than the potential energy at the extremes, i.e.

$$E < \lim_{x \rightarrow \pm\infty} V(x) \quad (2.1)$$

Scattering states are typically not normalizable solutions, and will have a continuous energy spectra. These are called as such because of the travelling wave solution one obtains when $V(x) = 0$ can be treated as scattering experiments. The energy spectra of this state must be greater than the potential energy at infinities.

$$E > \lim_{x \rightarrow \pm\infty} V(x) \quad (2.2)$$

2.0.2 Parity Operator

We introduce an operator called the parity operator, $\hat{\Pi}$. It has the effect of spatial inversion i.e.

$$\hat{\Pi}|\mathbf{r}\rangle = |-\mathbf{r}\rangle \quad (2.3)$$

It also has the effect of direction inversion for a momentum ket.

$$\hat{\Pi}|\mathbf{p}\rangle = |-\mathbf{p}\rangle \quad (2.4)$$

It is an involutory, hermitian and unitary operator, therefore:

$$\hat{\Pi} = \hat{\Pi}^\dagger = \hat{\Pi}^{-1} \quad (2.5)$$

Consider the eigenvalue equation of the operator.

$$\hat{\Pi}|\alpha\rangle = \lambda|\alpha\rangle \implies |\alpha\rangle = \lambda\hat{\Pi}|\alpha\rangle \quad (\because \hat{\Pi}^2 = \hat{I}) \quad (2.6)$$

From this, we can conclude that the eigenvalues of the parity operator are ± 1 . The eigenkets of the parity operator are called states of definite parity, or simply parity states.

There are both odd parity states $|\alpha_{-}\rangle$ with the eigenvalue -1 and even parity states $|\alpha_{+}\rangle$ with the eigenvalue 1 .

$$\text{Odd Parity} \implies \hat{\Pi}|\alpha_{-}\rangle = -|\alpha_{-}\rangle \quad (2.7)$$

$$\text{Even Parity} \implies \hat{\Pi}|\alpha_{+}\rangle = +|\alpha_{+}\rangle \quad (2.8)$$

Any arbitrary state can be written as a superposition of parity states, which we prove by considering the state $\frac{1}{2}(\hat{I} \pm \hat{\Pi})|\alpha\rangle$. Applying the parity operator would give us:

$$\hat{\Pi} \left(\frac{1}{2}(\hat{I} \pm \hat{\Pi})|\alpha\rangle \right) = \frac{1}{2}(\hat{\Pi}|\alpha\rangle \pm |\alpha\rangle) = \pm \left(\frac{1}{2}(\hat{I} \pm \hat{\Pi})|\alpha\rangle \right) \quad (2.9)$$

And therefore we can write the following:

$$|\alpha\rangle = |\alpha_{-}\rangle + |\alpha_{+}\rangle \quad (2.10)$$

$$\text{where } |\alpha_{-}\rangle = \frac{1}{2}(\hat{I} - \hat{\Pi})|\alpha\rangle \quad (2.11)$$

$$|\alpha_{+}\rangle = \frac{1}{2}(\hat{I} + \hat{\Pi})|\alpha\rangle \quad (2.12)$$

The parity operator being unitary gives us a unitary transform, we consider how it handles the position and momentum operators.

$$\langle \mathbf{r}' | \hat{\Pi} \hat{\mathbf{r}} \hat{\Pi} | \mathbf{r} \rangle = \langle -\mathbf{r}' | \hat{\mathbf{r}} | -\mathbf{r} \rangle = -\mathbf{r} \delta^3(\mathbf{r} - \mathbf{r}') = \langle \mathbf{r}' | (-\hat{\mathbf{r}}) | \mathbf{r} \rangle \quad (2.13)$$

$$\therefore \hat{\Pi} \hat{\mathbf{r}} \hat{\Pi} = -\hat{\mathbf{r}} \implies \hat{\Pi} \hat{\mathbf{r}} = -\hat{\mathbf{r}} \hat{\Pi} \quad (2.14)$$

Similarly, we get $\hat{\Pi} \hat{\mathbf{p}} \hat{\Pi} = -\hat{\mathbf{p}}$ and $\hat{\Pi} \hat{\mathbf{p}} = -\hat{\mathbf{p}} \hat{\Pi}$. Now, we consider the commutability of the parity and Hamiltonian operators.

$$[\hat{\Pi}, \hat{H}] = \hat{\Pi} \hat{H} - \hat{H} \hat{\Pi} = \hat{\Pi} \left(\frac{1}{2m} \hat{\mathbf{p}}^2 + V(\mathbf{r}) \right) - \left(\frac{1}{2m} \hat{\mathbf{p}}^2 + V(\mathbf{r}) \right) \hat{\Pi} \quad (2.15)$$

$$= \left(\frac{1}{2m} (-\hat{\mathbf{p}})^2 + V(-\mathbf{r}) \right) \hat{\Pi} - \left(\frac{1}{2m} \hat{p}^2 + V(\mathbf{r}) \right) \hat{\Pi} \quad (2.16)$$

$$= (V(-\mathbf{r}) - V(\mathbf{r})) \hat{\Pi} \quad (2.17)$$

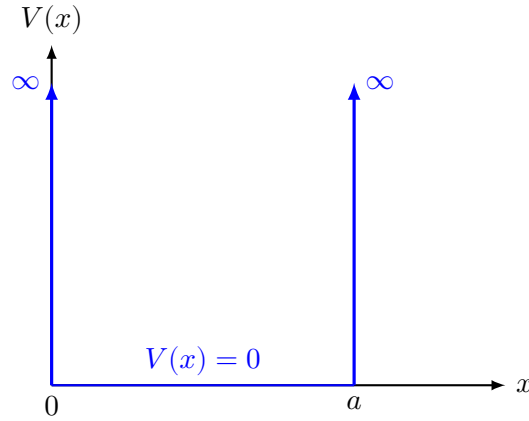
We see that the parity and Hamiltonian operators commute when the potential is symmetric/even i.e. $V(\mathbf{r}) = V(-\mathbf{r})$. This means that we have simultaneous eigenkets of energy and parity, which gives a very useful tool for even potentials.

$$\hat{\Pi}|n\rangle = \pm|n\rangle \implies \langle x | \hat{\Pi} | n \rangle = \pm \langle x | n \rangle \implies \langle -x | n \rangle = \pm \langle x | n \rangle \quad (2.18)$$

2.1 Infinite Square Well

We start with a very basic potential: a particle confined in a box, which is represented by the infinite square well potential.

$$V(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 < x < a \\ \infty, & x > a \end{cases} \quad (2.19)$$



We see that $E > V(\pm\infty)$, and therefore it is a bound state. Let us begin by constructing the Hamiltonian operator for this system.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V} \xrightarrow{\hat{V}=0} \frac{\hat{p}^2}{2m} \quad (2.20)$$

Now, we look for energy eigenkets of the system to find the energy spectrum.

$$\hat{H}|n\rangle = E_n|n\rangle \implies \frac{1}{2m}\hat{p}^2|n\rangle = E_n|n\rangle \implies \langle x|\hat{p}^2|n\rangle = 2mE_n\langle x|n\rangle \quad (2.21)$$

We use our result from Equation (1.143) to write $\hat{p}^2$ in position space.

$$\partial_x^2 \langle x|n\rangle = -\frac{2mE_n}{\hbar^2} \langle x|n\rangle \quad (2.22)$$

If we let $k = \sqrt{2mE_n}/\hbar$ (we will use this substitution a lot, so unless mentioned, k refers to this!), we solve the PDE to get:

$$\langle x|n\rangle = A \sin(kx) + B \cos(kx) \quad (0 < x < a) \quad (2.23)$$

The infinitely tall potential walls at $x = 0$ and $x = a$ dictate that the particle stays in the region $0 < x < a$, which gives us the boundary conditions that $\langle 0|n\rangle = \langle a|n\rangle = 0$. Applying the boundary conditions gives us:

$$A \sin(ka) + B \cos(ka) = 0 \quad ; \quad B = 0 \quad (2.24)$$

This implies that ka take integral values of π to satisfy the boundary conditions.

$$ka = \frac{a\sqrt{2mE_n}}{\hbar} = n\pi \quad (2.25)$$

This finally gives us the energy spectrum of the particle in a box.

$$\therefore E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} = n^2 E_1 \quad (2.26)$$

Therefore we should get $\langle x|n\rangle$ as following.

$$\langle x|n\rangle = A \sin\left(\frac{n\pi}{a}x\right) \quad (0 < x < a) \quad (2.27)$$

Now, to determine the constant A , we use the fact that $|n\rangle$ must be normalized, therefore:

$$1 = \langle n|n\rangle = \int_{-\infty}^{+\infty} \langle n|x\rangle \langle x|n\rangle dx \quad (2.28)$$

$$= \int_0^a A^2 \sin^2\left(\frac{n\pi}{a}x\right) dx = \frac{A^2}{2}a \quad (2.29)$$

$$\therefore A = \sqrt{\frac{2}{a}} \quad (2.30)$$

Putting this back where it belongs, we finally get that:

$$\langle x|n\rangle = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) \quad (0 < x < a) \quad (2.31)$$

And therefore, the energy eigenkets of the infinite square well are:

$$\therefore |n\rangle = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right) |x\rangle dx \quad (2.32)$$

Now, let's tackle the problem of time evolution of the state ket in this system. Since we have a time-independent Hamiltonian, we can write the time-evolution operator as:

$$\mathcal{U}(t) = e^{-i\hat{H}t/\hbar} \quad (2.33)$$

Therefore the ket expansion can be written as:

$$|\mathcal{S}(t)\rangle = \mathcal{U}(t)|\mathcal{S}\rangle = \sum_{n=1} |\mathcal{S}\rangle \langle n| e^{-i\hat{H}t/\hbar} |n\rangle \quad (2.34)$$

Now, from Equation (1.178) we can write further:

$$|\mathcal{S}(t)\rangle = \sum_{n=1} c_n e^{-iE_n t/\hbar} |n\rangle \quad (2.35)$$

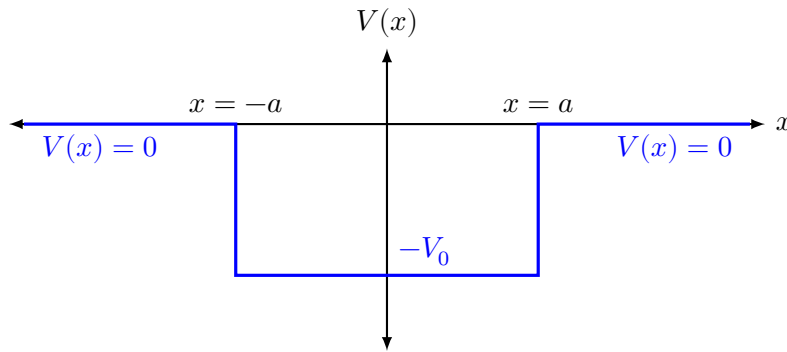
where $c_n = \langle n|\mathcal{S} \rangle$ are the expansion coefficients, given by:

$$c_n = \langle n|\mathcal{S} \rangle = \int_{-\infty}^{+\infty} \langle n|x \rangle \langle x|\mathcal{S} \rangle dx = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right) \langle x|\mathcal{S} \rangle dx \quad (2.36)$$

2.2 Finite Square Well

We once again consider a potential well, but this time the walls of the well are not infinitely tall, which allows the position to take values that would be restricted in the infinite square well.

$$V(x) = \begin{cases} 0, & |x| > a \\ -V_0, & |x| < a \end{cases} \quad V_0, a > 0 \quad (2.37)$$



Upon further inspection, we notice that this potential has both a bound state and a scattering state. We shall first consider the bound state with energy $E < 0$. The scattering state is covered in the final section of this chapter.

We once again begin by constructing the Hamiltonian and trying to find the energy eigenkets. In this case, we have two different values of potential, and therefore two different values of the Hamiltonian: $\hat{H} = \hat{p}^2/2m$ in the region $|x| < a$ and outside it, we have the $\hat{H} = \hat{p}^2/2m - V_0$. Lets tackle the region $0 < x < a$ first.

$$\hat{p}^2|n\rangle = 2mE_n|n\rangle \implies \partial_x^2 \langle x|n\rangle = -k^2 \langle x|n\rangle \quad (2.38)$$

Here, $k^2 = 2mE_n/\hbar^2$. The solution to this would be:

$$\langle x|n\rangle = A \sin kx + B \cos kx \quad (0 < x < a) \quad (2.39)$$

Now, for the region $x > a$ we have:

$$\left(\frac{1}{2m}\hat{p}^2 - V_0\right)|n\rangle = E_n|n\rangle \implies \partial_x^2 \langle x|n\rangle = -l^2 \langle x|n\rangle \quad (2.40)$$

Here, $l^2 = 2m(E_n + V_0)/\hbar^2$ and the solution to this would be:

$$\langle x|n\rangle = Ce^{lx} + De^{-lx} \quad (x > a) \quad (2.41)$$

Now, if you have noticed, we are dealing with a symmetric potential and therefore the energy

eigenkets are also eigenkets of the parity operator. So we have even or odd $\langle x|n\rangle$ functions.

Even Parity State

For even parity states, we have the following relations:

$$\langle x|n_+\rangle = \langle -x|n_+\rangle \quad (2.42)$$

$$x < -a \implies \langle x|n_+\rangle = Ae^{-kx} + Be^{kx} \implies A = B \quad (2.43)$$

$$-a < x < 0 \implies \langle x|n_+\rangle = -C \sin(lx) + D \cos(lx) \implies C = 0 \quad (2.44)$$

We also shave off the e^{-kx} term for $x > a$ and e^{kx} term for $x < -a$ to keep our kets normalizable. Therefore:

$$\langle x|n_+\rangle = \begin{cases} Ae^{kx}, & x < -a \\ D \cos(lx), & |x| < a \\ Ae^{-kx}, & x > a \end{cases} \quad (2.45)$$

We also have to preserve the continuity of $\langle x|n_+\rangle$ and its derivative, so we have the following relations:

$$Ae^{ka} = D \cos(la) \quad ; \quad -Ake^{ka} = -Dl \sin(la) \quad (2.46)$$

From which we get the following relation between k and l .

$$l \tan(la) = k \quad (2.47)$$

Now, using the definitions of l and k , we write that:

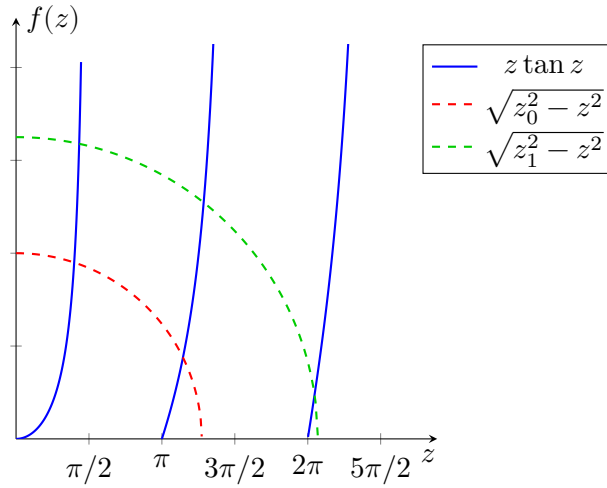
$$l^2 + k^2 = \frac{2mV_0}{\hbar^2} \quad (2.48)$$

Using this, we write:

$$(la) \tan(la) = \sqrt{\left(\frac{\sqrt{2mV_0a}}{\hbar}\right)^2 - (la)^2} \quad (2.49)$$

Let $z = la$ and $z_0^2 = 2mV_0a/\hbar^2$, then we get the following transcendental equation, whose solutions we obtain by numerical methods or drawing plots.

$$z \tan z = \sqrt{z_0^2 - z^2} \quad (2.50)$$



We see that for higher values of z_0 , we have more solutions to the even bound state. As $z_0 \rightarrow \infty$, we would see intersections at exactly odd multiples of $\pi/2$, which would give us the energy:

$$E_n + V_0 = \frac{n^2 \hbar^2 \pi^2}{8ma^2}, \quad n = 1, 3, 5, \dots \quad (2.51)$$

Odd Parity State

A similar analysis of the odd parity state would give us:

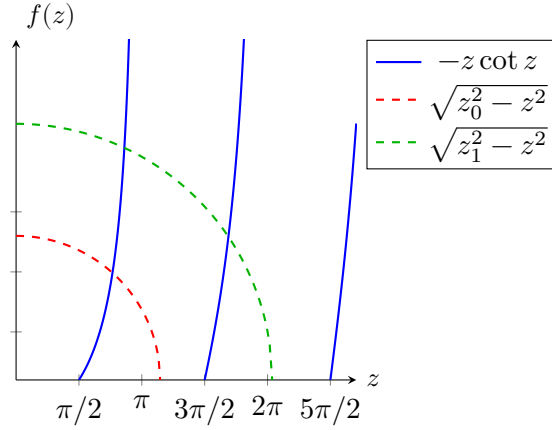
$$\langle x | n_- \rangle = \begin{cases} -Ae^{kx}, & x < -a \\ C \sin(lx), & |x| < a \\ Ae^{-kx}, & x > a \end{cases} \quad (2.52)$$

And the continuity condition gives the following equation:

$$-l \cot(la) = k \quad (2.53)$$

Which we rewrite by taking the same z and z_0 ,

$$-z \cot z = \sqrt{z_0^2 - z^2} \quad (2.54)$$



Observe that for $z_0 < \pi/2$, we would have no solutions to the odd bound state. There may always be an even bound state, but there may not be an odd one. And if we have $z_0 \rightarrow \infty$, we see the intersections at exactly the multiples of π , or we can say the even multiples of $\pi/2$. Therefore, the energy in this state would be:

$$E_n + V_0 = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 2, 4, 6, \dots \quad (2.55)$$

We see the energies distributed evenly between the odd and even states, with the even state getting odd multiples of n and vice-versa.

If you remember, we can write any ket as a sum of an odd and even parity ket, and we do this with the energy eigenkets. So we write:

$$|n\rangle = |n_+\rangle + |n_-\rangle \quad (2.56)$$

With the Hamiltonian being time-independent, the time-evolution operator is $e^{-i\hat{H}t/\hbar}$ and the state ket expansion of the system would be:

$$|\mathcal{S}(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |n\rangle \quad (2.57)$$

$$\text{where } c_n = \langle n | \mathcal{S} \rangle \quad (2.58)$$

2.3 Delta Function Well

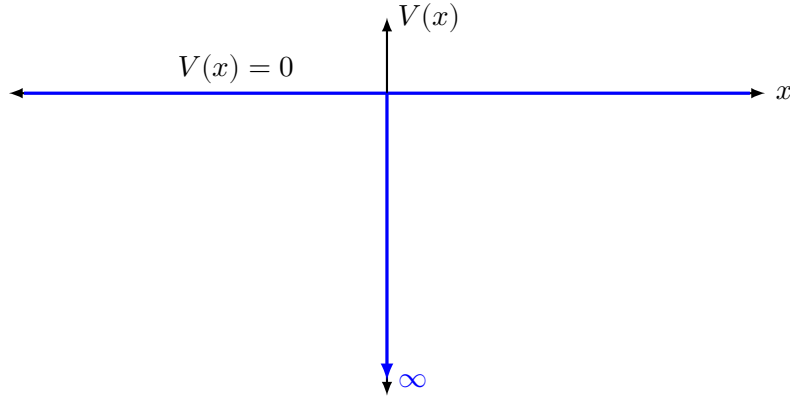
We now look towards analyzing a potential well described by Dirac's delta function, which is of the form:

$$V(x) = -\alpha \delta(x) \quad (2.59)$$

One should remember these two properties of the delta function.

$$\int_{-\infty}^{+\infty} \delta(x) dx = 1 \quad (2.60)$$

$$\int_{a-\epsilon}^{a+\epsilon} f(x) \delta(x-a) dx = f(a) \quad (\epsilon > 0) \quad (2.61)$$



Again, we isolate our discussion to only the bound states $E < 0$, and leave scattering states for later. The Hamiltonian for this potential is:

$$\hat{H} = \frac{1}{2m}\hat{p}^2 - \alpha\delta(\hat{x}) \xrightarrow{x \neq 0} \frac{1}{2m}\hat{p}^2 \quad (2.62)$$

$$\hat{p}^2|n\rangle = 2mE_n|n\rangle \implies \partial_x^2\langle x|n\rangle = k^2\langle x|n\rangle \quad (2.63)$$

Note that we have to divide our stationary state into two parts, since we have that anomaly at $x = 0$.

$$\langle x|n\rangle = \begin{cases} Ae^{kx} + Be^{-kx}, & x < 0 \\ Ce^{kx} + De^{-kx}, & x > 0 \end{cases} \quad (2.64)$$

We proceed by wiping off the e^{-kx} term for $x < 0$ and e^{kx} term to preserve the normalization of the stationary state, so $B = C = 0$ and we also get $A = D$ since we require the continuity of $\langle x|n\rangle$.

$$\langle x|n\rangle = \begin{cases} Ae^{kx}, & x < 0 \\ Ae^{-kx}, & x > 0 \end{cases} = Ae^{-k|x|} \quad (2.65)$$

Now, we also have to consider the continuity of the derivative of $\langle x|n\rangle$, which we do by considering the eigenvalue equation in position basis again.

$$-\frac{\hbar^2}{2m}\partial_x^2\langle x|n\rangle - \alpha\langle x|n\rangle\delta(x) = E_n\langle x|n\rangle \quad (2.66)$$

Now, for some $\epsilon > 0$, lets intergrate over the region $|x| < \epsilon$:

$$-\frac{\hbar^2}{2m}\int_{-\epsilon}^{+\epsilon}\partial_x^2\langle x|n\rangle dx - \int_{-\epsilon}^{+\epsilon}\alpha\langle x|n\rangle\delta(x) dx = \int_{-\epsilon}^{+\epsilon}E_n\langle x|n\rangle dx \quad (2.67)$$

From equation (2.61), we apply to get:

$$-\frac{\hbar^2}{2m}(\partial_x\langle x|n\rangle)\Big|_{-\epsilon}^{+\epsilon} - \alpha\langle 0|n\rangle = \int_{-\epsilon}^{+\epsilon}E_n\langle x|n\rangle dx \quad (2.68)$$

We take the limit as $\epsilon \rightarrow 0$, and we get the following result:

$$\Delta(\langle 0|n\rangle') = \frac{2m\alpha}{\hbar^2}\langle 0|n\rangle \quad (2.69)$$

Applying this relation, we get:

$$-Ake^{k \cdot 0} - Ake^{-k \cdot 0} = \frac{2m\alpha}{\hbar^2} Ae^{k|0|} \implies k = \frac{m\alpha}{\hbar^2} \quad (2.70)$$

$$\therefore E = -\frac{m\alpha^2}{2\hbar^2} \quad (2.71)$$

We therefore find that only a single bound state solution exists, and no others. Then:

$$\langle x|1\rangle = Ae^{-m\alpha|x|/\hbar^2} \quad (2.72)$$

Normalizing the stationary state would give us $A = \sqrt{m\alpha}/\hbar$, and therefore the state ket of the system would be written as follows:

$$|\mathcal{S}(t)\rangle = e^{-iEt/\hbar}|1\rangle = \frac{\sqrt{m\alpha}}{\hbar} \exp\left[-\frac{im\alpha^2 t}{2\hbar^3}\right] \int_{-\infty}^{+\infty} e^{-m\alpha|x|/\hbar^2} |x\rangle dx \quad (2.73)$$

2.4 Harmonic Oscillator

The harmonic oscillator is one of the most important potentials to analyze, since an important application is that one can approximate any well as a harmonic oscillator at the appropriate limit using a power series expansion.

$$\hat{V}(\hat{x}) = \frac{1}{2}m\omega^2\hat{x}^2 \quad (2.74)$$

2.4.1 Power Series Solution

We once again begin by writing the energy eigenvalue equation and moving into position space.

$$\hat{H} = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2 \quad (2.75)$$

$$-\frac{\hbar^2}{2m}\partial_x^2\langle x|n\rangle + \frac{1}{2}m\omega^2x^2\langle x|n\rangle = E_n\langle x|n\rangle \quad (2.76)$$

First, we write the differential equation in terms of $\xi = x\sqrt{M\omega/\hbar}$ and $\epsilon = 2E_n/\hbar\omega$ and for convenience, $\langle \xi|n\rangle = u(\xi)$.

$$u''(\xi) + (\epsilon - \xi^2)u(\xi) = 0 \quad (2.77)$$

Now, we expect $u(\xi)$ to asymptote to zero at the infinities, otherwise it would not be normalizable. For this to be true for all ϵ , $u(\xi)$ should be of the form:

$$u(\xi) = h(\xi)f(\xi) \quad (2.78)$$

and it should solve the differential equation:

$$f''(\xi) - \xi^2 f(\xi) = 0 \implies f(\xi) = e^{-\xi^2/2} \quad (2.79)$$

$$u(\xi) = h(\xi)e^{-\xi^2/2} \quad (2.80)$$

and from this, $h(\xi)$ should solve the following differential equation:

$$h''(\xi) - 2\xi h'(\xi) + (2\epsilon - 1)h(\xi) = 0 \quad (2.81)$$

We proceed by assuming a power series solution to $h(\xi)$

$$h(\xi) = \sum_{n=0}^{\infty} a_n \xi^n \quad (2.82)$$

$$\sum_{n=0}^{\infty} n(n-1)a_n \xi^{n-2} - 2na_n \xi^n + (\epsilon - 1)a_n \xi^n = 0 \quad (2.83)$$

and for the ξ^{n-2} term, we shift from $n \rightarrow n+2$ so that we have a common factor of ξ^n , and this should not affect the sum since the first two terms are zero anyway.

$$\sum_{n=0}^{\infty} [(n+1)(n+2)a_{n+2} + (\epsilon - 2n - 1)a_n] \xi^n = 0 \quad (2.84)$$

$\{\xi^n\}$ forms a linearly independent basis, so for the linear combination to be zero, the coefficients must be zero. We therefore obtain a recursion relation.

$$a_{n+2} = \frac{2n+1-\epsilon}{(n+1)(n+2)} a_n \quad (2.85)$$

Therefore, for we obtain a unique a_0 and a_1 , we obtain a unique solution. We further write the function $h(\xi)$ as a sum of its even and odd powered terms:

$$h(\xi) = h_+(\xi) + h_-(\xi) \quad (2.86)$$

One point to be noted is that not all specified a_0, a_1 will give normalizable solutions. This must mean we need a finite number of terms for normalizability. If one term becomes zero, then the rest also become zero, so for $n \geq 0$ then $a_{n+2} = 0$ and we have the relation $\epsilon = 2n + 1$, and finally, we obtain the energies:

$$\therefore E_n = \left(n + \frac{1}{2}\right) \hbar\omega \quad (2.87)$$

For allowed values of ϵ , the recursion relation then becomes:

$$a_{j+2} = \frac{2(j-n)}{(j+1)(j+2)} a_j \quad (2.88)$$

One must note that for allowed values of ϵ , it can only truncate either the odd or even part of $h(\xi)$, so the other part will be totally zero. These are the *Hermite polynomials*, represented by

$H_n(\xi)$. The ξ^n term of the Hermite polynomials have the coefficient 2^n by convention. Using the recursion formula, we get

$$\langle \xi|0\rangle = a_0 e^{-\xi^2/2} \quad (2.89)$$

$$\langle \xi|1\rangle = a_1 \xi e^{-\xi^2/2} \quad (2.90)$$

$$\langle \xi|2\rangle = a_0(1 - \xi^2)e^{-\xi^2/2} \quad (2.91)$$

After normalization, we finally get the ξ basis energy eigenket to be:

$$\langle \xi|n\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n \cdot n!}} H_n(\xi) e^{-\xi^2/2} \quad (2.92)$$

2.4.2 Dirac's Solution

This algebraic solution to the harmonic oscillator was first proposed by P.A.M. Dirac. The Hamiltonian for this situation is given by:

$$\hat{H} = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2 = \frac{1}{2M}[\hat{p}^2 + (m\omega\hat{x})^2] \quad (2.93)$$

It is convenient to define two operators: the annihilation operator $\mathcal{A}$ and the creation operator $\mathcal{A}^\dagger$.

$$\mathcal{A} = \frac{1}{\sqrt{2m\hbar\omega}}(i\hat{p} + m\omega\hat{x}) \quad \mathcal{A}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}}(-i\hat{p} + m\omega\hat{x}) \quad (2.94)$$

Using the canonical commutation relations (1.111), we obtain the commutator.

$$[\mathcal{A}, \mathcal{A}^\dagger] = \hat{I} \quad (2.95)$$

and we further define the number operator $\mathcal{N} = \mathcal{A}^\dagger \mathcal{A}$.

$$\mathcal{N} = \mathcal{A}^\dagger \mathcal{A} = \frac{1}{2m\hbar\omega}(i\hat{p} + m\omega\hat{x})(i\hat{p} - m\omega\hat{x}) \quad (2.96)$$

$$= \frac{1}{2M\hbar\omega}(\hat{p}^2 + (m\omega\hat{x})^2 - im\omega[\hat{x}, \hat{p}]) \quad (2.97)$$

$$= \frac{1}{\hbar\omega} \left(\frac{1}{2m}\hat{p}^2 + m\omega\hat{x}^2 - \frac{1}{2}\hbar\omega \right) = \frac{1}{\hbar\omega}\hat{H} - \frac{1}{2} \quad (2.98)$$

$$\therefore \hat{H} = \hbar\omega \left(\mathcal{N} + \frac{1}{2} \right) \quad (2.99)$$

Therefore, we can see that $[\hat{H}, \mathcal{N}] = 0$, therefore they will have simultaneous eigenkets $|n\rangle$, with eigenvalues n .

$$\mathcal{N}|n\rangle = n|n\rangle \quad (2.100)$$

which would imply that the energy spectrum of the harmonic oscillator is therefore:

$$\therefore E_n = \hbar\omega \left(n + \frac{1}{2} \right) \quad (2.101)$$

We need $E_n > 0$ for this state to be normalizable, so this means $n \geq 0$. We continue by finding the commutators of $\mathcal{N}$ with $\mathcal{A}$ and $\mathcal{A}^\dagger$.

$$[\mathcal{N}, \mathcal{A}] = [\mathcal{A}^\dagger \mathcal{A}, \mathcal{A}] = -\mathcal{A} \quad (2.102)$$

$$[\mathcal{N}, \mathcal{A}^\dagger] = [\mathcal{A}^\dagger \mathcal{A}, \mathcal{A}^\dagger] = \mathcal{A}^\dagger \quad (2.103)$$

As a result, we can derive that:

$$\mathcal{N} \mathcal{A}^\dagger |n\rangle = ([\mathcal{N}, \mathcal{A}^\dagger] + \mathcal{A}^\dagger \mathcal{N}) |n\rangle = (n+1) \mathcal{A}^\dagger |n\rangle \quad (2.104)$$

$$\mathcal{N} \mathcal{A} |n\rangle = ([\mathcal{N}, \mathcal{A}] - \mathcal{A} \mathcal{N}) |n\rangle = (n-1) \mathcal{A} |n\rangle \quad (2.105)$$

These relations imply that $\mathcal{A}^\dagger |n\rangle$ ($\mathcal{A} |n\rangle$) is an eigenket of $\mathcal{N}$ with the eigenvalue $n+1$ ($n-1$). We interpret this as $\mathcal{A}^\dagger$ ($\mathcal{A}$) operating on $|n\rangle$ would increase (decrease) the energy of the eigenket by $\hbar\omega$, hence the names creation (annihilation) operator. Further, this would imply $\mathcal{A}^\dagger$ ($\mathcal{A}$) is changing the state from $|n\rangle$ to $|n+1\rangle$ ($|n-1\rangle$), and therefore we can write the following equations:

$$\mathcal{A} |n\rangle = c |n-1\rangle \quad (2.106)$$

$$\Rightarrow \langle n | \mathcal{A}^\dagger = c^* \langle n-1 | \quad (2.107)$$

$$\langle n | \mathcal{A}^\dagger \mathcal{A} |n\rangle = \langle n | \mathcal{N} |n\rangle = |c|^2 \quad (2.108)$$

And by properties of the number operator, we see $c = \sqrt{n}$, if we take c to be real and positive by convention. Performing a similar thing with $\mathcal{A}^\dagger$, we get the two relations:

$$\mathcal{A} |n\rangle = \sqrt{n} |n-1\rangle \quad (2.109)$$

$$\mathcal{A}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \quad (2.110)$$

Since $n \geq 0$, an application of $\mathcal{A}$ to an arbitrary $|n\rangle$ n times would give us the lowest (ground) state $|0\rangle$ with the energy $E_0 = \hbar\omega/2$. We now apply the creation operator to the ground state to finally get the energy eigenkets.

$$|1\rangle = \mathcal{A}^\dagger |0\rangle \quad (2.111)$$

$$|2\rangle = \left(\frac{\mathcal{A}^\dagger}{\sqrt{2}} \right) |1\rangle = \frac{1}{\sqrt{2!}} (\mathcal{A}^\dagger)^2 |0\rangle \quad (2.112)$$

$$|3\rangle = \left(\frac{\mathcal{A}^\dagger}{\sqrt{3}} \right) |2\rangle = \frac{1}{\sqrt{3!}} (\mathcal{A}^\dagger)^3 |0\rangle \quad (2.113)$$

$$\vdots \quad (2.114)$$

$$|n\rangle = \frac{1}{\sqrt{n!}} (\mathcal{A}^\dagger)^n |0\rangle \quad (2.115)$$

Now, to find the ground state ket, we know there is no state less than the ground state, so the

annihilation operator should make the state null, and we move into the position basis.

$$\mathcal{A}|0\rangle = 0 \implies \langle x|\mathcal{A}|0\rangle = 0 \quad (2.116)$$

$$\langle x|\left[\frac{1}{\sqrt{2m\hbar\omega}}(i\hat{p} + m\omega\hat{x})\right]|0\rangle = 0 \implies \partial_x \langle x|0\rangle = -\frac{m\omega}{\hbar}x\langle x|0\rangle \quad (2.117)$$

The solution to this differential equation, after normalization is:

$$\langle x|0\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar} \quad (2.118)$$

We often take $\xi = x\sqrt{m\omega/\hbar}$, $\xi' = \sqrt{m\omega/\hbar}$ and $\alpha = (m\omega/\pi\hbar)^{1/4}$ to simplify calculations. The excited state can be obtained by evaluating $\langle x|(\mathcal{A}^\dagger)^n|0\rangle$, which in general we get:

$$\langle x|n\rangle = \frac{\alpha}{\sqrt{2^n \cdot n!}}(\xi')^{n+1/2} \left(x - \frac{1}{(\xi')^2}D_x\right)^n \langle x|0\rangle = \frac{\alpha}{\sqrt{n!}}\langle x|(\mathcal{A}^\dagger)^n|0\rangle \quad (2.119)$$

2.5 Free Particle and Wave Packets

2.5.1 Free Particle

We start by considering the potential $V(x) = 0$, for all values of x . You will see why this isn't the first potential we analyzed. The Hamiltonian for this potential would be $\hat{H} = \hat{p}^2/2m$ and the associated position space differential equation is:

$$\partial_x^2 \langle x|n\rangle = -k^2 \langle x|n\rangle \quad (2.120)$$

whose solution is the following:

$$\langle x|n\rangle = Ae^{ikx} + Be^{-ikx} \quad (2.121)$$

e^{ikx} and e^{-ikx} represent the plane waves travelling in the $+x$ and $-x$ direction. I'd like to move to using wave functions for this part. We further separate the wave functions by the plane wave direction:

$$\psi_\pm(x) = A_\pm e^{\pm ikx} \quad (2.122)$$

We add on the standard time dependence factor $e^{-iE_n t/\hbar}$ for stationary states, and we write $E_n = \hbar^2 k^2/2m$.

$$\Psi_\pm(x, t) = A_\pm e^{-i[(\hbar k^2/2m)t \mp kx]} \quad (2.123)$$

There are few subtleties to this solution, we shall discuss them now. First, the probability density is constant everywhere.

$$|\Psi_\pm(x, t)|^2 = |A_\pm|^2 \quad (2.124)$$

Second, we discuss velocities. A classical wave has the velocity ω/k , and a classical particle should have a velocity p/m . Since this wave function has to represent such a particle, we expect to see $v_{\text{wave}} = v_{\text{particle}}$. The angular frequency ω is given by Planck–Einstein relation $E = \hbar\omega$ and to find the momentum, we go back to the energy eigenvalue equation and move into momentum

space.

$$\hat{p}^2|n\rangle = \hbar^2 k^2|n\rangle \implies p^2\langle p|n\rangle = \hbar^2 k^2\langle p|n\rangle \quad (2.125)$$

$$\therefore p = \pm \hbar k \quad (2.126)$$

So therefore, we have momenta $\pm \hbar k$ associated with the wave functions $\Psi_{\pm}(x, t)$. Now, the wave and particle velocities can be calculated:

$$v_{\text{wave}} = \frac{\omega}{k} = \frac{\hbar k}{2m} \quad (2.127)$$

$$v_{\text{particle}} = \frac{p}{m} = \frac{\hbar k}{m} \quad (2.128)$$

$$v_{\text{wave}} = 2v_{\text{particle}} \quad (2.129)$$

Another point to be noted, is if we notice the group velocity of the wave, it matches the particle velocity.

$$v_{\text{group}} = D_k \omega(k) = D_k \left(\frac{\hbar k^2}{2m} \right) = \frac{\hbar k}{m} = v_{\text{particle}} \quad (2.130)$$

Third, we notice the solutions are not normalizable. The $\pm x$ solutions blow up to infinity while $x \rightarrow \mp \infty$.

These problems actually have a pretty neat solution. We consider superpositions of the waves at extremes, such that the wave function asymptotes to zero. We essentially attempt to localize the wave function by this superposition. This localized wave is called a *wave packet*, made up of groups of waves that have varying wavelengths/frequencies that interact constructively and destructively.

2.5.2 Wave Packets

Localized wave packets can be constructed by superposing, in the same region of space, waves of slightly different wavelengths, but with phases and amplitudes chosen to make the superposition constructive in the desired region and destructive outside it. Mathematically, we can carry out this superposition by means of *Fourier transforms*.

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{i(kx - \omega t)} dx \quad (2.131)$$

and to find $\phi(k)$, the initial state of the wave function should be mentioned, and through Plancherel's theorem:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \Psi(x, 0) e^{-ikx} dx \quad (2.132)$$

$$\Psi(x, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{ikx} dx \quad (2.133)$$

The particle is represented not by a single de Broglie wave of well-defined frequency and wavelength, but by a wave packet that is obtained by adding a large number of waves of different

frequencies. This is why we see a discrepancy in wave and particle velocities, it is the group of waves that represents the particle. Using $k = p/\hbar$, one can also recover the momentum space transformation.

$$\Phi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \Psi(x, t) e^{-ipx/\hbar} dx \quad (2.134)$$

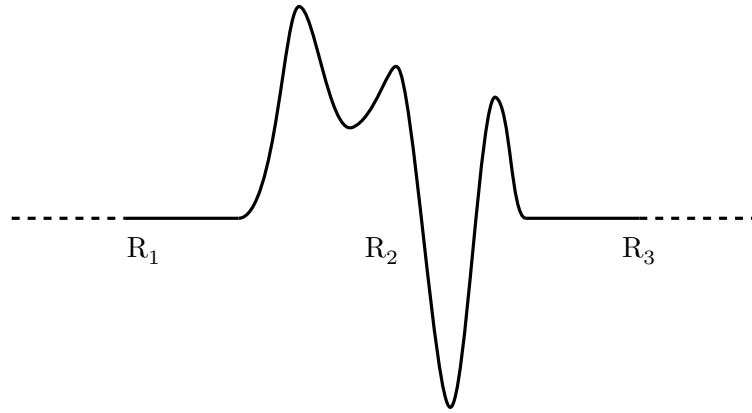
2.6 Scattering States

2.6.1 Transmission and Reflection

A scattering state is a continuous, non-normalizable state in quantum mechanics corresponding to a particle with an energy E greater than the asymptotic values of the potential energy function $V(x)$ as $x \rightarrow \pm\infty$. Mathematically, it is a stationary state solution in the position basis which remains oscillatory at infinity. A typical scattering state's solution will be of the following form.

$$\langle x|n\rangle = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x \in R_1 \\ Cf(x) + Dg(x), & x \in R_2 \\ Fe^{ikx} + Ge^{-ikx}, & x \in R_3 \end{cases} \quad (2.135)$$

where regions R_1 and R_3 have a constant potential V_0 , the energy of the system is $E > V_0$, and the region R_2 has potential $V(x \in R_2) < V_0$, and where $k = \sqrt{2m(V_0 - E)}/\hbar$.



For simplicity, we take V_0 to be zero. The earlier section tells us that stationary states in the position basis of the form $Ae^{\pm ikx}$ represent a free particle moving in the $\pm x$ direction. There, the potential is zero throughout, we however have regions of both zero and non-zero potential. Remember Rutherford's α -scattering experiment? There, the α -particle (which are free) hit the gold nuclei and are scattered, with some being "reflected" and some are "transmitted" through the atom. The scattering process involves three stages: the first and third being "free" stages, where no forces are involved and the second stage is a where the force interaction happens. We can see an analogue here, with the regions $|x| \geq a$ being the "free" regions and the region $|x| < a$ as the interaction region. Some particles are transmitted and some are reflected, and we shall quantify them using the ratios of the probability flux.

$$T = \frac{j_t}{j_i} \quad R = \frac{j_r}{j_i} \quad (2.136)$$

T and R are the transmission and reflection coefficients and j_t, j_r and j_i are the transmitted, reflected and incident probability fluxes, respectively. Since T and R are probabilities that the particle is transmitted and reflected, they should add up to unity.

$$T + R = 1 \quad (2.137)$$

Scattering can happen from one direction, in one dimension we can have from the left to right or right to left. So there should not be any wave-particle in the right direction for right to left scattering and vice-versa. We first consider left-right scattering, and we should have $G = 0$. The probability fluxes are:

$$j_i = \frac{i\hbar}{2m}(\psi^* \partial_x \psi - \psi \partial_x \psi^*) \quad (2.138)$$

$$= \frac{i\hbar}{2m}(A^* e^{-ikx} \cdot k A e^{ikx} - A e^{ikx} \cdot (-A^* k e^{-ikx})) \quad (2.139)$$

$$= \frac{i\hbar}{2m}(2k|A|^2) = \frac{i\hbar k}{m}|A|^2 \quad (2.140)$$

$$\text{Similarly, } j_r = \frac{i\hbar k}{m}|B|^2 \quad (2.141)$$

$$j_t = \frac{i\hbar k}{m}|F|^2 \quad (2.142)$$

and the transmission and reflection coefficients are:

$$T_r = \frac{|F|^2}{|A|^2} \quad R_r = \frac{|B|^2}{|A|^2} \quad (2.143)$$

In a right-left scattering experiment, we would have $A = 0$ and the transmission and reflection coefficients would be:

$$T_l = \frac{|B|^2}{|G|^2} \quad R_l = \frac{|F|^2}{|G|^2} \quad (2.144)$$

One can exploit the symmetry of potentials, for an even potential, we can show that $T_r = T_l$ and $R_r = R_l$.

2.6.2 Scattering and Transfer Matrices

We begin constructing two vectors χ_{in} and χ_{out} , which house the incoming and outgoing matter waves.

$$\chi_{\text{in}} = \begin{pmatrix} A \\ G \end{pmatrix} \quad \chi_{\text{out}} = \begin{pmatrix} B \\ F \end{pmatrix} \quad (2.145)$$

The total probability flux in regions R_1 and R_3 would be:

$$j_1 = \frac{i\hbar}{m}(|A|^2 - |B|^2) \quad (2.146)$$

$$j_3 = \frac{i\hbar}{m}(|F|^2 - |G|^2) \quad (2.147)$$

The should be equal, since probability needs to be conserved.

$$|A|^2 + |G|^2 = |B|^2 + |F|^2 \quad (2.148)$$

We can rewrite this in the form of the vectors χ_{in} and χ_{out} .

$$\chi_{\text{in}}^\dagger \chi_{\text{in}} = \chi_{\text{out}}^\dagger \chi_{\text{out}} \quad (2.149)$$

This implies the existence of a unitary matrix that transforms the incoming into the outgoing, which we call the *scattering matrix*, or S -matrix for short. Since it is unitary, we have $S^\dagger S = \hat{I}$.

$$\chi_{\text{out}} = S\chi_{\text{in}} \iff \begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix} \quad (2.150)$$

The components of the S -matrix are given by the right and left transmission and reflection amplitudes.

$$S = \begin{pmatrix} \left. \frac{B}{A} \right|_{G=0} & \left. \frac{F}{A} \right|_{G=0} \\ \left. \frac{F}{G} \right|_{A=0} & \left. \frac{B}{G} \right|_{A=0} \end{pmatrix} = \begin{pmatrix} r_r & t_r \\ t_l & r_l \end{pmatrix} \quad (2.151)$$

Where $R = |r|^2$ and $T = |t|^2$. In a similar fashion, we define two vectors χ_1 and χ_3 that house the matter waves in the regions R_1 and R_3 , and using the conservation of probability flux, we have:

$$|A|^2 - |B|^2 = |F|^2 - |G|^2 \quad (2.152)$$

We further use a matrix Ω to write that:

$$\Omega = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (2.153)$$

$$\chi_1^\dagger \Omega \chi_1 = \chi_3^\dagger \Omega \chi_3 \quad (2.154)$$

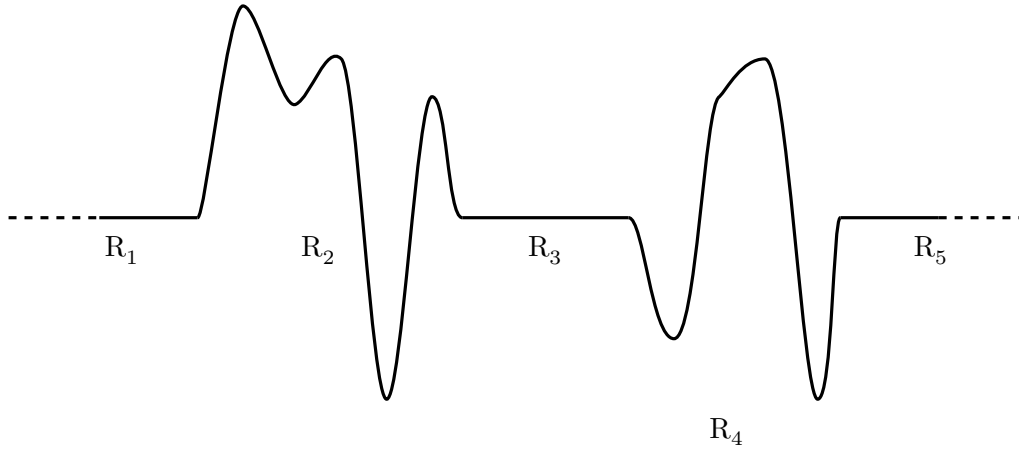
We further define a matrix M , called the *transfer matrix*, which relate the coefficients A and B to F and G . It also has the property $M^\dagger \Omega M = \hat{I}$.

$$\chi_3 = M\chi_1 \iff \begin{pmatrix} F \\ G \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} \quad (2.155)$$

S-Matrix and M-Matrix Relation

$$S = \frac{1}{M_{22}} \begin{pmatrix} -M_{21} & 1 \\ \det(M) & M_{12} \end{pmatrix} \quad (2.156)$$

$$M = \frac{1}{S_{21}} \begin{pmatrix} 1 & -S_{22} \\ S_{11} & -\det(S) \end{pmatrix} \quad (2.157)$$



A property of the M -matrix is that if we have multiple regions where we have a plane wave solution, then the M -matrix for the entire thing would be the product of each M -matrix. Mathematically, if the M -matrix for boundary R_2 and R_4 is M_1 and M_2 , then the transfer matrix for the entire boundary $R_2 + R_3 + R_4$ would be:

$$M = M_1 M_2 \quad (2.158)$$

This can be extended upto any number of boundaries, and the proof for this is trivial.

2.6.3 Finite Square Well

In a scattering state, we remember that $E > V(\pm\infty)$ and in this case $E > 0$. So we once again, begin by finding the energy eigenkets. In the region $|x| > a$, we have the Hamiltonian $\hat{p}^2/2m$ and the eigenvalue equation:

$$\hat{p}^2|n\rangle = 2mE_n|n\rangle \implies \partial_x^2\langle x|n\rangle = -k^2\langle x|n\rangle \quad (2.159)$$

And for the region $|x| < a$, we have $\hat{H} = \hat{p}^2/2m - V_0$ and the eigenvalue equation:

$$\hat{p}^2|n\rangle = 2m(E_n + V_0)|n\rangle \implies \partial_x^2\langle x|n\rangle = -l^2\langle x|n\rangle \quad (2.160)$$

The overall solution will be the following:

$$\langle x|n\rangle = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < -a \\ Ce^{ilx} + De^{-ilx}, & |x| < a \\ Fe^{ikx} + Ge^{-ikx}, & x > a \end{cases} \quad (2.161)$$

The continuity and derivative continuity requirement would further give us the following relations:

$$Ae^{-ika} + Be^{ika} = Ce^{-ila} + De^{ila} ; Ce^{ila} + De^{-ila} = Fe^{ika} + Ge^{-ika} \quad (2.162)$$

$$k(Ae^{-ika} - Be^{ika}) = l(Ce^{-ila} - De^{ila}) ; l(Ce^{ila} - De^{-ila}) = k(Fe^{ika} - Ge^{-ika}) \quad (2.163)$$

Since the potential is symmetric, we have $T_l = T_r$. Using these relations, the transmission coefficient is given by $|F/A|^2$:

$$\frac{1}{T} = 1 + \frac{V_0}{4E(E + V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right) \quad (2.164)$$

The transmission and reflection amplitudes are:

$$r = \frac{i \frac{l^2 - k^2}{2kl} \sin(2la)}{\cos(2la) - i \frac{k^2 + l^2}{2kl} \sin(2la)} \quad (2.165)$$

$$t = \frac{e^{-2ika}}{\cos(2qa) - i \frac{k^2 + q^2}{2kq} \sin(2qa)} \quad (2.166)$$

The scattering and transfer matrices are therefore given by:

$$S = \begin{pmatrix} r & t \\ t & r \end{pmatrix} \quad M = \frac{1}{t} \begin{pmatrix} 1 & -r \\ r & t^2 - r^2 \end{pmatrix} \quad (2.167)$$

2.6.4 Delta Function Well

With the Hamiltonian $\hat{H} = \hat{p}^2/2m$ for $x \neq 0$ and $E > 0$, we get the following energy eigenkets in position space:

$$\langle x|n \rangle = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0 \\ Fe^{ikx} + Ge^{-ikx}, & x > 0 \end{cases} \quad (2.168)$$

And the continuity conditions give us the following relations:

$$A + B = F + G \quad (2.169)$$

$$F - G = A(1 + 2i\beta) - B(1 - 2i\beta), \quad \text{where } \beta \equiv \frac{m\alpha}{\hbar^2 k} \quad (2.170)$$

The symmetry of the problem lets us say the left and right scattering experiments will be identical, and the reflection and transmission amplitudes are given by:

$$r = \frac{i\beta}{1 - i\beta} \quad t = \frac{1}{1 - i\beta} \quad (2.171)$$

From which we get the reflection and transmission coefficients:

$$R = \frac{\beta^2}{1 + \beta^2} \quad T = \frac{1}{1 + \beta^2} \quad (2.172)$$

It is trivial to get the scattering and transfer matrices from this.

2.6.5 Potential Step

We consider a new potential, the potential step. Our task is to analyze the scattering state $E > V_0$.

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x > 0 \end{cases} \quad (2.173)$$

In the region $x < 0$, the Hamiltonian is $\hat{H} = \hat{p}^2/2m$ and we get the stationary state:

$$\langle x|n\rangle = Ae^{ikx} + Be^{-ikx} \quad (x < 0) \quad (2.174)$$

and in the region $x > 0$, the Hamiltonian is $\hat{H} = \hat{p}^2/2m - V_0$ and the energy eigenstate is:

$$\langle x|n\rangle = Fe^{ilx} + Ge^{-ilx} \quad (x > 0) \quad (2.175)$$

where we have $l = \sqrt{2m(E - V_0)}/\hbar$. We exploit the continuity of the eigenstate and its derivative and we get that:

$$A + B = F + G \quad (2.176)$$

$$k(A - B) = l(F - G) \quad (2.177)$$

You should have noticed by now that the two regions have differing wave numbers k and l , and we cannot just say $|F/A|^2$ is the transmission coefficient. We must return to the original definition in terms of probability flux to get an accurate answer.

$$j_t = \frac{i\hbar}{2m} (Fe^{ilx} \cdot lF^*e^{-ilx} - F^*e^{-ilx} \cdot -lFe^{-ilx}) \quad (2.178)$$

$$= \frac{i\hbar l}{m} |F|^2 \quad (2.179)$$

$$\text{Similarly, } j_i = \frac{i\hbar k}{m} |A|^2 \quad (2.180)$$

$$j_r = \frac{i\hbar k}{m} |B|^2 \quad (2.181)$$

$$T_r = \frac{j_t}{j_i} = \frac{l}{k} \cdot \frac{|F|^2}{|A|^2} \quad R_r = \frac{j_r}{j_i} = \frac{|B|^2}{|A|^2} \quad (2.182)$$

Using the relations of continuity we obtained earlier, it is easy to get that:

$$T = \frac{4lk}{(l+k)^2} \quad R = \left(\frac{l-k}{l+k} \right)^2 \quad (2.183)$$

3 HYDROGEN AND ANGULAR MOMENTUM

The atom is something physicists have tried uncover for centuries, with various theoretical models developed by the likes of Thomson, Rutherford and Bohr. With access to the Schrödinger Equation, we can now definitively answer what goes on in the atom. Along with that, we go for an detailed analysis of Angular Momentum and its role as a generator of rotations.

3.1 Hydrogen Atom

3.1.1 Radial Equation

We start by formulating the Schrodinger Wave Equation in Spherical Coordinates. The cartesian version is given by:

$$i\hbar\partial_t\Psi(\mathbf{r},t) = -\frac{\hbar^2}{2m}\nabla^2\Psi(\mathbf{r},t) + V(\mathbf{r},t)\Psi(\mathbf{r},t)$$

We look for solutions to a time independent potential, then for $\Psi(\mathbf{r},t) = \psi(\mathbf{r})e^{-iEt/\hbar}$,

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

We begin by writing the Laplacian in its spherical form, here the tilde represents the unit vector.

$$\nabla = \partial_r\tilde{\mathbf{r}} + \frac{1}{r}\partial_\theta\tilde{\theta} + \frac{1}{r\sin\theta}\partial_\phi\tilde{\phi} \quad (3.1)$$

$$\nabla^2 = \frac{1}{r^2}\partial_r(r^2\partial_r) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta) + \frac{1}{r^2\sin^2\theta}\partial_\phi^2 \quad (3.2)$$

And we furthur assume a seperable solution of $\psi(\mathbf{r}) = R(r)Y(\theta, \phi)$ to a central potential $V(r)$ and we write the Schrodinger equation as:

$$-\frac{\hbar^2}{2m}\left[\frac{Y}{r^2}\partial_r(r^2\partial_r R) + \frac{R}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta Y) + \frac{R}{r^2\sin^2\theta}\partial_\phi^2 Y\right] + RYV = RYE \quad (3.3)$$

$$\left\{\frac{1}{R}\partial_r(r^2\partial_r R) - \frac{2mr^2}{\hbar^2}(V(r) - E)\right\} = -\frac{1}{Y}\left\{\frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta Y) + \frac{1}{\sin^2\theta}\partial_\phi^2 Y\right\} \quad (3.4)$$

They are equal if each is seperately equal to a constant, say $\ell(\ell+1)$, and we consider the Radial part here.

$$\partial_r(r^2\partial_r R) - \frac{2mr^2}{\hbar^2}(V(r) - E)R = \ell(\ell+1)R \quad (3.5)$$

We consider a substitution $u(r) \equiv rR(r)$, then we obtain the *Radial Equation*.

$$-\frac{\hbar^2}{2m}\partial_r^2 u + \left[V + \frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2}\right]u = Eu \quad (3.6)$$

This is quite similar to the TISE aside from the *centrifugal term* $\hbar^2\ell(\ell+1)/2mr^2$.

3.1.2 Spherical Harmonics

We approach the angular part of the equation. We begin by again forcing separable solutions $Y(\theta, \phi) = \Theta(\theta)\Phi(\phi)$.

$$\sin \theta \partial_\theta (\sin \theta \partial_\theta Y) + \partial_\phi^2 Y = -\ell(\ell + 1) \sin^2 \theta Y \quad (3.7)$$

$$\left\{ \frac{1}{\Theta} [\sin \theta \partial_\theta (\sin \theta \partial_\theta \Theta)] + \ell(\ell + 1) \sin^2 \theta \right\} = \frac{1}{\Phi} \partial^2 \Phi(\phi) \quad (3.8)$$

We take the separation constant to be m_ℓ^2 , and get:

$$\left\{ \frac{1}{\Theta} [\sin \theta \partial_\theta (\sin \theta \partial_\theta \Theta)] + \ell(\ell + 1) \sin^2 \theta \right\} = \frac{1}{\Phi} \partial^2 \Phi(\phi) \quad (3.9)$$

We also know that (r, θ, ϕ) and $(r, \theta, \phi + 2\pi)$ are the exact same point, so the wave function at those points must be equal.

$$e^{im_\ell \phi} e^{i2\pi m_\ell} = e^{im_\ell \phi} \implies m_\ell = 0, \pm 1, \pm 2, \dots \quad (3.10)$$

Therefore we find m_ℓ to be an integer, and we move onto the Θ part.

$$\sin \theta \partial_\theta (\sin \theta \partial_\theta \Theta) + [\ell(\ell + 1) \sin^2 \theta - m_\ell^2] \Theta = 0 \quad (3.11)$$

If we force ℓ to be an integer, we get the solution to be the associated Legendre function in $\cos \theta$.

$$\Theta(\theta) = A P_\ell^{m_\ell}(\cos \theta) \quad (3.12)$$

$$P_\ell^{m_\ell}(x) = (-1)^{m_\ell} (1 - x^2)^{m_\ell/2} D_x^{m_\ell} P_\ell(x) \quad (3.13)$$

where $P_\ell(x)$ is the ℓ 'th Legendre polynomial, whose Rodrigues formula is:

$$P_\ell(x) = \frac{1}{2^\ell \ell!} D_x^\ell (x^2 - 1)^\ell \quad (3.14)$$

and together with the ϕ part, the normalized angular wave functions are called the *spherical harmonics*.

$$Y_\ell^{m_\ell}(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m_\ell)!}{(\ell + m_\ell)!}} e^{im_\ell \phi} P_\ell^{m_\ell}(\cos \theta) \quad (3.15)$$

with ℓ and m_ℓ taking the following values:

$$\ell = 0, 1, 2, \dots \implies m_\ell = -\ell, -\ell + 1, \dots, -1, 0, 1, \dots, \ell - 1, \ell$$

The spherical harmonics also have the following property:

$$Y_\ell^{-m_\ell} = (-1)^{m_\ell} (Y_\ell^{m_\ell})^*$$

The spherical harmonics are also the eigenfunctions of angular momentum, as we'll see later.

3.1.3 Hydrogen Atom

Now that we have the tools to analyze central potentials, we can finally take a look at the simplest atomic system: the hydrogen atom. The potential here is the Coulomb electrostatic potential between the electron and proton.

$$V(r) = -\frac{1}{4\pi\epsilon_0} \frac{e}{r} \quad (3.16)$$

We use the Radial equation for this potential, since we already know the spherical harmonics for a given ℓ and m_ℓ .

$$-\frac{\hbar^2}{2m_e} \partial_r^2 u + \left[-\frac{e}{4\pi\epsilon_0 r} + \frac{\hbar^2}{2m_e} \frac{\ell(\ell+1)}{r^2} \right] u = Eu \quad (3.17)$$

We simplify this by using $\kappa = \sqrt{-2m_e E}/\hbar$. $\rho = \kappa r$ and $\rho_0 = m_e e^2 / 2\pi\epsilon_0 \hbar^2 \kappa$, we get:

$$\partial_\rho^2 u = \left[1 - \frac{\rho_0}{\rho} + \frac{\ell(\ell+1)}{\rho^2} \right] u \quad (3.18)$$

And we further take another substitution $u(\rho) = \rho^{\ell+1} e^{-\rho} v(\rho)$ to get:

$$\rho \partial_\rho^2 v + 2(\ell+1-\rho) \partial_\rho v + (\rho_0 - 2(\ell+1))v = 0 \quad (3.19)$$

Such an equation can be solved using a power series to get the recursion formula:

$$c_{j+1} = \frac{2(j+\ell+1) - \rho_0}{(j+1)(j+2\ell+2)} c_j \quad (3.20)$$

This series blows up, so we need it to terminate for some c_N , and therefore:

$$\rho_0 = 2(N+\ell) \equiv 2n \quad (3.21)$$

We use ρ_0 to now get the energies of the Hydrogen atom to be:

$$E_n = - \left[\frac{m_e e^4}{32\hbar^2 \pi^2 \epsilon_0^2} \right] \frac{1}{n^2} = \frac{E_1}{n^2}, \quad n = 1, 2, 3, \dots \quad (3.22)$$

and here E_1 is calculated to be -13.6 eV. This is the famous *Bohr formula* for the Hydrogen atom's spectrum, which was calculated before Wave Mechanics was formalised. We also find that:

$$\kappa = \left(\frac{m_e e^2}{4\pi\epsilon_0 \hbar^2} \right) \frac{1}{n} = \frac{1}{an} \quad (3.23)$$

$$a \equiv \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = 0.529 \times 10^{-10} \text{m} \quad (3.24)$$

This the *Bohr radius*, also calculated along with Bohr formula. From this we rewrite ρ as:

$$\rho = \frac{r}{an} \quad (3.25)$$

So we finally get our normalized Radial wave functions to be:

$$R_{n\ell}(r) = \sqrt{\left(\frac{2}{an}\right)^3 \frac{(n-\ell-1)!}{2n \cdot (n+\ell)!}} e^{-r/na} \left(\frac{2r}{an}\right)^\ell \cdot L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{an}\right) \quad (3.26)$$

where $L_q^p(x)$ and $L_q(x)$ are the associated Laguerre polynomial and q th Laguerre polynomial respectively.

$$L_q^p(x) = (-1)^p D_x^p L_{p+q}(x) \quad (3.27)$$

$$L_q(x) = \frac{e^x}{q!} D_x^q (e^{-x} x^q) \quad (3.28)$$

Note that due to this, ℓ can take the values $1, 2, 3, \dots, n-1$. This gives the Hydrogen atom a degeneracy of $d_n = n^2$, not including the spin of the electron.

3.2 Angular Momentum

3.2.1 Rotation Commutation

We proceed with analysis of rotation of kets. We use $R_z(\delta)$ to represent an infinitesimal rotation about an axis k . Using the rotation matrices about each axis, we write:

$$R_x(\delta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 - \delta^2/2 & -\delta \\ 0 & \delta & 1 - \delta^2/2 \end{pmatrix} \quad (3.29)$$

$$R_y(\delta) = \begin{pmatrix} 1 - \delta^2/2 & 0 & \delta \\ 0 & 1 & 0 \\ -\delta & 0 & 1 - \delta^2/2 \end{pmatrix} \quad (3.30)$$

$$R_z(\delta) = \begin{pmatrix} 1 - \delta^2/2 & -\delta & 0 \\ \delta & 1 - \delta^2/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.31)$$

Let's see if the rotations about different axes commutes.

$$[R_x(\delta), R_y(\delta)] = \begin{pmatrix} 0 & -\delta^2 & 0 \\ \delta^2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R_z(\delta^2) - 1 \quad (3.32)$$

We see that rotations about different axes *do not commute* unlike translations. Using the Levi-Civita symbol and the summation convention, we generalize it as:

$$[R_j(\delta), R_k(\delta)] = \epsilon_{jkl} R_l(\delta) - 1 \quad (3.33)$$

3.2.2 Rotation Operator

We now talk about the rotation operator $\mathcal{D}(R)$, where R represents any rotation operation.

$$\mathcal{D}(R)|\alpha\rangle = |\alpha\rangle_R \quad (3.34)$$

By a similar analysis to the infinitesimal translation and time evolution operators, we can write the infinitesimal rotation operator as such:

$$\mathcal{D}_k(\delta) = 1 - i\hat{G}_k\delta \quad (3.35)$$

We finally postulate the operator $\hat{G}$ to be proportional to the generalized angular momentum operator $\hat{J}_k$. Therefore an infinitesimal rotation about k -axis is given by:

$$\mathcal{D}_k(\delta) = 1 - i\hat{J}_k\delta/\hbar \quad (3.36)$$

and a more general rotation about some unit vector $\mathbf{n}$ would be:

$$\mathcal{D}_{\mathbf{n}}(\delta) = 1 - i(\hat{\mathbf{J}} \cdot \mathbf{n})\delta/\hbar \quad (3.37)$$

It is important to note that angular momentum is not defined to be $\hat{\mathbf{r}} \times \hat{\mathbf{p}}$ and this is due to the existence of *spin* angular momentum. Namely,

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \quad (3.38)$$

where $\mathbf{L}$ is the classical, orbital angular momentum defined to be $\mathbf{r} \times \mathbf{p}$ and $\mathbf{S}$ is the spin angular momentum, which is a purely quantum mechanical entity. Now, for a finite rotation, we just apply an the infinitesimal one for a large N times.

$$\mathcal{D}_k(\theta) = \lim_{N \rightarrow \infty} \left[1 - i\hat{J}_k \left(\frac{\theta}{N} \right) / \hbar \right]^N = \exp \left[-\frac{i\hat{J}_k\theta}{\hbar} \right] \quad (3.39)$$

Using the rotation commutation relation we derived earlier, we write that:

$$(1 - i\hat{J}_x\delta/\hbar)(1 - i\hat{J}_y\delta/\hbar) - (1 - i\hat{J}_y\delta/\hbar)(1 - i\hat{J}_x\delta/\hbar) = 1 - i\hat{J}_z\delta/\hbar - 1$$

$$\hat{J}_x\hat{J}_y - \hat{J}_y\hat{J}_x = i\hbar\hat{J}_z$$

or, more generally:

$$[\hat{J}_j, \hat{J}_k] = i\hbar\epsilon_{jkl}\hat{J}_l \quad (3.40)$$

Just like the operators $\mathcal{T}$ and $\mathcal{U}$, $\mathcal{D}$ is unitary, associative and has an inverse, which is its hermitian conjugate. However, it is not commutative for rotations about different axes. We regard the generalized angular momentum as the $\mathbf{J}$ to be the generator of rotations, similar to the way we declared momentum and the Hamiltonian to be generators of translation and time evolution, respectively. Before we talk about the eigenkets of $\mathbf{J}$, lets talk about orbital and spin angular momenta and their generalization first.

3.2.3 Generalized Angular Momentum

Orbital angular momentum derived using classical properties. Using the classical angular momentum definition,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \quad (3.41)$$

it is easy to find that the orbital angular momentum about each axis is given by:

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \quad (3.42)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \quad (3.43)$$

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \quad (3.44)$$

or more generally:

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \quad (3.45)$$

And using the canonical commutation relation, we find the commutation relations to be:

$$[\hat{L}_j, \hat{L}_k] = i\hbar\epsilon_{jkl}\hat{L}_l \quad (3.46)$$

Also note that if we define $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$, we find that all the components do indeed commute with $\hat{L}^2$.

$$[\hat{L}^2, \hat{\mathbf{L}}] = 0 \quad (3.47)$$

This means that $\hat{L}^2, \hat{L}_x, \hat{L}_y$ and $\hat{L}_z$ are a complete set of commuting observables, and we have a common eigenket for each of them.

However, in 1922, the Stern–Gerlach experiment accidentally discovered Spin. The setup involved passing neutral Ag atoms through an inhomogenous magnetic and detecting their final position. Typically, one would expect to find a smear of atoms on the detector, but instead what they found was two distinct spots where the atoms had clumped up, and in equal proportions. The explanation to this was given by Goudsmit and Uhlenbeck 3 years later, who proposed the concept of spin angular momentum. Spin is a uniquely quantum observable, and it has no classical analogue. Sometimes it is said that spin can be thought of as the particle itself spinning, however that notion is quite misleading. Spin angular momentum has the same properties as its orbital counterpart.

$$[\hat{S}_j, \hat{S}_k] = i\hbar\epsilon_{jkl}\hat{S}_l \quad (3.48)$$

$$[\hat{S}^2, \hat{\mathbf{S}}] = 0 \quad (3.49)$$

This similarity led to the development of a generalized angular momentum quantity, represented by J , which is the sum of the orbital and spin angular momenta.

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \quad (3.50)$$

It follows all the same properties of L and S , since it is just a sum of those two. Note that we can do this despite the basis of $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$ being differential operators and matrix mechanics, two seemingly different things. This is because the basis of $\hat{\mathbf{J}}$ is the group theory, to be specific, lie groups and algebras. The generalized angular momentum is what determines the rotation completely in quantum mechanics.

3.2.4 Eigensystem of Angular Momentum

Let's now look to finding the eigenvectors and eigenvalues of angular momentum. We know that $\hat{J}_x, \hat{J}_y$ and $\hat{J}_z$ form a complete set of commuting observables with $\hat{J}^2$, so we look to diagonalize a pair. By convention, we take $\hat{J}_z$ and $\hat{J}^2$. Writing out their eigenvalue equations, by taking α and β to be dimensionless constants,

$$\hat{\mathbf{J}}^2 |\alpha, \beta\rangle = \alpha \hbar^2 |\alpha, \beta\rangle \quad (3.51)$$

$$\hat{J}_z |\alpha, \beta\rangle = \beta \hbar |\alpha, \beta\rangle \quad (3.52)$$

We further define two operators $\hat{J}_\pm$ such that,

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y \quad (3.53)$$

from which we reconstruct $\hat{J}_x$ and $\hat{J}_y$ as:

$$\hat{J}_x = \frac{1}{2} (\hat{J}_+ + \hat{J}_-) \quad (3.54)$$

$$\hat{J}_y = \frac{1}{2i} (\hat{J}_+ - \hat{J}_-) \quad (3.55)$$

and we also find that:

$$\hat{J}_x^2 = \frac{1}{4} (\hat{J}_+^2 + \hat{J}_-^2 + \hat{J}_+ \hat{J}_- + \hat{J}_- \hat{J}_+) \quad (3.56)$$

$$\hat{J}_y^2 = -\frac{1}{4} (\hat{J}_+^2 + \hat{J}_-^2 - \hat{J}_+ \hat{J}_- - \hat{J}_- \hat{J}_+) \quad (3.57)$$

from which we obtain the following commutation relations:

$$[\hat{\mathbf{J}}^2, \hat{J}_\pm] = 0 \quad (3.58)$$

$$[\hat{J}_+, \hat{J}_-] = 2\hbar \hat{J}_z \quad (3.59)$$

$$[\hat{J}_z, \hat{J}_\pm] = \pm \hbar \hat{J}_\pm \quad (3.60)$$

In addition, $\hat{J}_+$ and $\hat{J}_-$ also satisfy,

$$\hat{J}_\pm \hat{J}_\mp = \hat{J}_x^2 + \hat{J}_y^2 \pm \hbar \hat{J}_z = \hat{J}^2 - \hat{J}_z^2 \pm \hbar \hat{J}_z \quad (3.61)$$

which gives us:

$$\hat{\mathbf{J}}^2 = \frac{1}{2} (\hat{J}_+ \hat{J}_- + \hat{J}_- \hat{J}_+) + \hat{J}_z^2 \quad (3.62)$$

Now, we apply the $\hat{J}_\pm$ to $|\alpha, \beta\rangle$ to see their effect. Using our relations we just found, we write:

$$\hat{J}_z (\hat{J}_\pm |\alpha, \beta\rangle) = (\hat{J}_\pm \hat{J}_z \pm \hbar \hat{J}_\pm) |\alpha, \beta\rangle = \hbar(\beta \pm 1) \hat{J}_\pm |\alpha, \beta\rangle \quad (3.63)$$

Which implies that $\hat{J}_+$ ($\hat{J}_-$) raises (lowers) the eigenvalue of J_z by a unit of $\hbar$. Therefore, it is called the raising (lowering) operator. Together, $\hat{J}_+$ and $\hat{J}_-$ are called *ladder operators*.

$$\hat{J}_\pm |\alpha, \beta\rangle = C_{\alpha, \beta}^\pm |\alpha, \beta \pm 1\rangle \quad (3.64)$$

And now for $\hat{J}^2$,

$$\hat{J}^2 (\hat{J}_\pm |\alpha, \beta\rangle) = \hat{J}_\pm (\hat{\mathbf{J}}^2 |\alpha, \beta\rangle) = \alpha \hbar^2 (\hat{J}_\pm |\alpha, \beta\rangle) \quad (3.65)$$

It seems like the raising and lowering do not affect the eigenvalue $\alpha \hbar^2$, so the sum of squares is always constant. Therefore, since $\hat{\mathbf{J}}^2 - \hat{J}_z^2$ is always positive, we can write:

$$\langle \alpha, \beta | \hat{J}^2 - \hat{J}_z^2 | \alpha, \beta \rangle = \hbar^2 (\alpha - \beta^2) \geq 0 \implies \alpha \geq \beta^2 \quad (3.66)$$

Since we see β has an upper limit, there must exist $|\alpha, \beta_{\max}\rangle$ that cannot be further raised.

$$\hat{J}_+ |\alpha, \beta_{\max}\rangle = 0 \implies \hat{J}_- \hat{J}_+ |\alpha, \beta_{\max}\rangle = 0 \quad (3.67)$$

$$(\hat{\mathbf{J}}^2 - \hat{J}_z^2 - \hbar \hat{J}_z) |\alpha, \beta_{\max}\rangle = 0 \implies \alpha = \beta_{\max}(\beta_{\max} + 1) \quad (3.68)$$

We also see there is a lower limit $\beta_{\min}$, which corresponds to a state that cannot be lowered further.

$$\hat{J}_- |\alpha, \beta_{\min}\rangle = 0 \implies \hat{J}_+ \hat{J}_- |\alpha, \beta_{\min}\rangle = 0 \quad (3.69)$$

$$(\hat{\mathbf{J}}^2 - \hat{J}_z^2 + \hbar \hat{J}_z) |\alpha, \beta_{\min}\rangle = 0 \implies \alpha = \beta_{\min}(\beta_{\min} - 1) \quad (3.70)$$

Comparing the values for α , we conclude that:

$$\beta_{\min} = -\beta_{\max} \quad (3.71)$$

If we let $|\alpha, \beta_{\min}\rangle = \hat{J}_-^n |\alpha, \beta_{\max}\rangle$, then $\beta_{\max} = \beta_{\min} + n$, then we conclude that $\beta_{\max} = -\beta_{\min} = n/2$. Here we introduce notation $j = \beta_{\max}$ and $m = \beta = -j, -j+1, \dots, j-1, j$ respectively. Now, we find the raising and lowering factors $C_{j,m}^\pm$.

$$\hat{J}_\pm |j, m\rangle = C_{j,m}^\pm |j, m \pm 1\rangle \quad (3.72)$$

One should notice that $\hat{J}_+$ and $\hat{J}_-$ are hermitian conjugates of each other, and we use that by,

$$(\hat{J}_{\pm}|j, m\rangle)^{\dagger}(\hat{J}_{\pm}|j, m\rangle) = \langle j, m|\hat{J}_{\mp}\hat{J}_{\pm}|j, m\rangle = |C_{j,m}^{\pm}|^2 \quad (3.73)$$

$$C_{j,m}^{\pm} = \sqrt{\langle j, m|\hat{J}^2 - \hat{J}_z^2 \mp \hbar\hat{J}_z|j, m\rangle} = \hbar\sqrt{j(j+1) - m(m \pm 1)} \quad (3.74)$$

and therefore, we finally write the eigenvalue equations as:

$$\hat{\mathbf{J}}^2|j, m\rangle = j(j+1)\hbar^2|j, m\rangle \quad (3.75)$$

$$\hat{J}_z|j, m\rangle = m\hbar|j, m\rangle \quad (3.76)$$

$$\hat{J}_{\pm}|j, m\rangle = \hbar\sqrt{j(j+1) - m(m \pm 1)}|j, m \pm 1\rangle \quad (3.77)$$

$$\hat{J}_x|j, m\rangle = \frac{\hbar}{2} \left[\sqrt{j(j+1) - m(m+1)}|j, m+1\rangle + \sqrt{j(j+1) - m(m-1)}|j, m-1\rangle \right] \quad (3.78)$$

$$\hat{J}_y|j, m\rangle = \frac{\hbar}{2i} \left[\sqrt{j(j+1) - m(m+1)}|j, m+1\rangle - \sqrt{j(j+1) - m(m-1)}|j, m-1\rangle \right] \quad (3.79)$$

3.2.5 Spin States and Spin 1/2

Spin follows the same rules that generalized and orbital angular momentum follow. The eigenvalue equations for spin are:

$$\hat{\mathbf{S}}^2|s, m_s\rangle = s(s+1)\hbar^2|s, m_s\rangle \quad (3.80)$$

$$\hat{S}_z|s, m_s\rangle = m_s\hbar|s, m_s\rangle \quad (3.81)$$

$$\hat{S}_x|s, m_s\rangle = \frac{\hbar}{2} \left[\sqrt{s(s+1) - m_s(m_s+1)}|s, m_s+1\rangle + \sqrt{s(s+1) - m_s(m_s-1)}|s, m_s-1\rangle \right] \quad (3.82)$$

$$\hat{S}_y|s, m_s\rangle = \frac{\hbar}{2i} \left[\sqrt{s(s+1) - m_s(m_s+1)}|s, m_s+1\rangle - \sqrt{s(s+1) - m_s(m_s-1)}|s, m_s-1\rangle \right] \quad (3.83)$$

$$\hat{S}_{\pm}|s, m_s\rangle = \hbar\sqrt{s(s+1) - m_s(m_s \pm 1)}|s, m_s \pm 1\rangle \quad (3.84)$$

It is important to discuss spin states, since we have to include it with the total state of the particle, with the total state being a tensor product of energy eigenket with the spin state.

$$|\psi\rangle = |n\rangle \otimes |s, m_s\rangle \quad (3.85)$$

The most fundamental state is that of $s = 1/2$, from which it is easy to get the properties of higher spin states. It is simplest state, since there are only two of them, $|1/2, 1/2\rangle$ and $|1/2, -1/2\rangle$, called *spin up* and *spin down* states. We typically just use $|\chi_{\pm}\rangle$ to represent them.

$$|\chi_{+}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\chi_{-}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (3.86)$$

The eigenvalue equation for spin 1/2 are:

$$\hat{\mathbf{S}}^2|\chi_{\pm}\rangle = \frac{3}{4}\hbar^2|\chi_{\pm}\rangle \quad (3.87)$$

$$\hat{S}_z|\chi_{\pm}\rangle = \pm\frac{\hbar}{2}|\chi_{\pm}\rangle \quad (3.88)$$

$$\hat{S}_x|\chi_{\pm}\rangle = \frac{\sqrt{3}}{4}\hbar|\chi_{\mp}\rangle \quad (3.89)$$

$$\hat{S}_y|\chi_{\pm}\rangle = \pm i\frac{\sqrt{3}}{4}\hbar|\chi_{\mp}\rangle \quad (3.90)$$

$$\hat{S}_{\pm}|\chi_{\pm}\rangle = \frac{\sqrt{3}}{4}\hbar|\chi_{\mp}\rangle \quad (3.91)$$

The matrix representation in the basis of $|\chi_{\pm}\rangle$ for is easy to get. For $\hat{\mathbf{S}}^2$ and $\hat{S}_z$, $|\chi_{\pm}\rangle$ are its eigenvectors and hence they would be diagonalized, with the diagonal elements being the eigenvalues.

$$\hat{\mathbf{S}}^2 = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (3.92)$$

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.93)$$

The matrix elements of $\hat{S}_x$ and $\hat{S}_y$ are given by:

$$(S_x)_{nm} = \langle\chi_n|\hat{S}_x|\chi_m\rangle \quad (3.94)$$

$$(S_y)_{nm} = \langle\chi_n|\hat{S}_y|\chi_m\rangle \quad (3.95)$$

From which it is trivial to get the matrices:

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (3.96)$$

$$\hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (3.97)$$

We can further simplify these matrices by removing the factor of $\hbar/2$, and the corresponding matrices are called the *Pauli spin matrices*.

Pauli Spin Matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (3.98)$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (3.99)$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.100)$$

The properties of the spin matrices can be derived from the spin operators, which are:

$$[\sigma_j, \sigma_k] = 2i\epsilon_{jkl}\sigma_l \quad (3.101)$$

$$\{\sigma_j, \sigma_k\} = 2\delta_{jk} \quad (3.102)$$

$$\sigma_x\sigma_y\sigma_z = 1 \quad (3.103)$$

$$e^{i\alpha\sigma_j} = \cos \alpha + i\sigma_j \sin \alpha \quad (3.104)$$

Writing out the eigenvalue equations for $\hat{S}_x$ and $\hat{S}_y$, we can get their eigenkets.

$$\hat{S}_x|\xi_{\pm}\rangle = \pm\frac{\hbar}{2}|\xi_{\pm}\rangle \quad (3.105)$$

$$\hat{S}_y|\gamma_{\pm}\rangle = \pm\frac{\hbar}{2}|\gamma_{\pm}\rangle \quad (3.106)$$

Solving it in the $|\chi_{\pm}\rangle$ basis, it is easy to get the the eigenvectors to be:

$$|\xi_{\pm}\rangle = \frac{1}{\sqrt{2}}[|\chi_{+}\rangle \pm |\chi_{-}\rangle] \quad (3.107)$$

$$|\gamma_{\pm}\rangle = \frac{1}{\sqrt{2}}[|\chi_{+}\rangle \pm i|\chi_{-}\rangle] \quad (3.108)$$

This means that diagonalization of S_z forces a superposition of the spin in the x and y directions, which means for one definite spin state, the other two will have some uncertainty. This is reinforced by the uncertainty principle, which gives:

$$\sigma_{S_x}\sigma_{S_y} \geq \frac{1}{2}|\langle S_z \rangle| \quad (3.109)$$

3.2.6 Eigenfunctions of Orbital Angular Momentum

The eigenvalue equations of orbital angular momentum are as follows:

$$\hat{\mathbf{L}}^2|\ell, m_{\ell}\rangle = \ell(\ell+1)\hbar^2|\ell, m_{\ell}\rangle \quad (3.110)$$

$$\hat{L}_z|\ell, m_{\ell}\rangle = m_{\ell}\hbar|\ell, m_{\ell}\rangle \quad (3.111)$$

Since we define $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, we can write it as differential operators in position space.

$$\langle \mathbf{r} | \hat{L}_z = \langle \mathbf{r} | (\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) = -i\hbar(x\partial_y - y\partial_x) \langle \mathbf{r} | \quad (3.112)$$

It is convenient for us to work in spherical coordinates for this problem. In that case, we have:

$$\langle r, \theta, \phi | \hat{L}_z = -i\hbar \partial_\phi \langle r, \theta, \phi | \quad (3.113)$$

We do the same for $\hat{L}_x$ and $\hat{L}_y$ to find $\hat{\mathbf{L}}^2$, which we get to be:

$$\langle r, \theta, \phi | \hat{\mathbf{L}}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2 \right] \langle r, \theta, \phi | \quad (3.114)$$

Since we try to find functions that are both eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$, let the function be $Y_\ell^{m_\ell}(\theta, \phi)$ (the differential operators are independent of the radial coordinate).

$$\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta Y_\ell^{m_\ell}) + \frac{1}{\sin^2 \theta} \partial_\phi^2 Y_\ell^{m_\ell} = -\ell(\ell+1) Y_\ell^{m_\ell} \quad (3.115)$$

$$-i\hbar \partial_\phi Y_\ell^{m_\ell} = Y_\ell^{m_\ell} \quad (3.116)$$

We recognize this to be the differential equations of the spherical harmonics.

$$Y_\ell^{m_\ell}(\theta, \phi) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m_\ell)!}{(\ell+m_\ell)!}} e^{im_\ell \phi} P_\ell^{m_\ell}(\cos \theta) \quad (3.117)$$

3.2.7 Addition of Angular Momentum

In this section we present the general formalism corresponding to the problem of adding two commuting angular momenta. The general angular momentum relations between we give as:

$$[\hat{J}_k^{(1)}, \hat{J}_l^{(2)}] = 0 \quad (3.118)$$

$$[\hat{J}_k^{(1)/(2)}, \hat{J}_l^{(1)/(2)}] = i\hbar \epsilon_{kl n} \hat{J}_n^{(1)/(2)} \quad (3.119)$$

We denote the eigenkets as $|j_1, m_1\rangle$ nad $|j_2, m_2\rangle$, and the eigenvalue equations would be:

$$(\hat{\mathbf{J}}^{(1)})^2 |j_1, m_1\rangle = j_1(j_1+1)\hbar^2 |j_1, m_1\rangle \quad (3.120)$$

$$(\hat{\mathbf{J}}^{(2)})^2 |j_2, m_2\rangle = j_2(j_2+1)\hbar^2 |j_2, m_2\rangle \quad (3.121)$$

$$\hat{J}_z^{(1)} |j_1, m_1\rangle = m_1 \hbar |j_1, m_1\rangle \quad (3.122)$$

$$\hat{J}_z^{(2)} |j_2, m_2\rangle = m_2 \hbar |j_2, m_2\rangle \quad (3.123)$$

The dimensions of the first and second angular momentum spaces are $2j_1+1$ and $2j_2+1$. Addition involves us to use simultaneous eigenkets of both spaces, which we get by a tensor/direct product $|j_1, m_1\rangle |j_2, m_2\rangle = |j_1, m_1, j_2, m_2\rangle$. The problem of adding two angular momenta is:

$$\hat{\mathbf{J}} = \hat{\mathbf{J}}^{(1)} + \hat{\mathbf{J}}^{(2)}$$

Adding the corresponding components, it is easy to recover the original angular momentum commutation relations.

$$[\hat{J}_k, \hat{J}_l] = i\hbar\epsilon_{klm}\hat{J}_m \quad (3.124)$$

And we find $\hat{\mathbf{J}}^2$ to be:

$$\hat{\mathbf{J}}^2 = (\hat{\mathbf{J}}^{(1)})^2 + (\hat{\mathbf{J}}^{(2)})^2 + 2\hat{J}_z^{(1)}\hat{J}_z^{(2)} + \hat{J}_+^{(1)}\hat{J}_+^{(2)} + \hat{J}_-^{(1)}\hat{J}_-^{(2)} \quad (3.125)$$

From this we find $(\hat{\mathbf{J}}^{(1)})^2$, $(\hat{\mathbf{J}}^{(2)})^2$, $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ to be commute with each other. This means we can designate $|j, m\rangle$ to be eigenkets of all four.

$$(\hat{\mathbf{J}}^{(1)})^2|j, m\rangle = j_1(j_1 + 1)|j, m\rangle \quad (3.126)$$

$$(\hat{\mathbf{J}}^{(2)})^2|j, m\rangle = j_2(j_2 + 1)|j, m\rangle \quad (3.127)$$

$$\hat{\mathbf{J}}^2|j, m\rangle = j(j + 1)|j, m\rangle \quad (3.128)$$

$$\hat{J}_z|j, m\rangle = m\hbar|j, m\rangle \quad (3.129)$$

The main problem of addition of angular momenta is to find the addition state $|j, m\rangle$ in terms of $|j_1, m_1\rangle|j_2, m_2\rangle$. We can insert an identity and write the following:

$$|j, m\rangle = \sum_{m_1, m_2} \langle j_1, m_1, j_2, m_2 | j, m \rangle |j_1, m_1, j_2, m_2\rangle \quad (3.130)$$

The coefficients $\langle j_1, m_1, j_2, m_2 | j, m \rangle$ are called the *Clebsh–Gordan coefficients*. They are taken to be real by convention. They can be looked up in a Clebsh–Gordan table, or be calculated by hand if you're a madman.

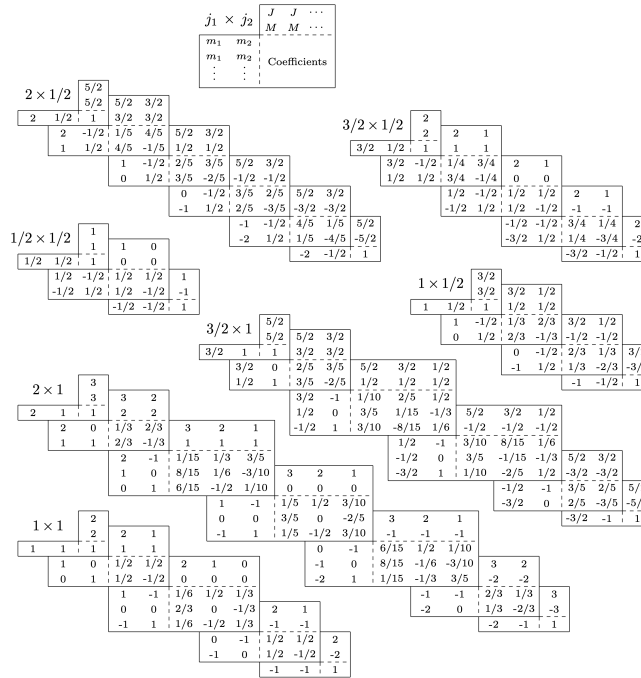


Figure 1: Clebsh-Gordan coefficients. A square root is understood on each coefficient, that is, $-1/3$ means $-\sqrt{1/3}$.